

Can co-folding confidence tell interacting IDR pairs apart?

Stage 1 of the IDR-IDR benchmark: 60 of 62 AlphaFold Server jobs scored, five samples each. The pre-registered primary metric is ipTM of the top-ranked model and the pre-registered comparator is a leave-one-out logistic model on sequence composition alone - not chance. A test passes only if the paired-bootstrap CI on delta-AUC excludes zero.

THE ANSWER, ON THESE 60 JOBS

No - ipTM does not beat sequence composition.

Every pre-registered test fails its own criterion, and in the homotypic arm AF3 is significantly worse than the trivial baseline.

JOBS SCORED

60 / 62

S20, S21 pending

TEST 1 DELTA AUC

-0.099

[-0.28, +0.08] FAIL

TEST 2B "THE RESULT"

-0.236

[-0.38, -0.08] FAIL

HOMOTYPIC ARM

-0.339

CI excludes zero: worse

GCN4 CANARY P21

0.73

pipeline behaves

The pre-registered tests primary metric ipTM; comparator = composition baseline

test / group	n pos/neg	AF3 AUC	95% CI	baseline	delta	delta 95% CI	verdict
Test 1 17 positives vs 31 real negatives	29/43	0.593	[0.46, 0.72]	0.692	-0.099	[-0.28, +0.08]	● FAIL
Test 2a mutual folding vs hard zipper	4/6	0.354	[0.00, 0.75]	0.333	+0.021	[-0.54, +0.58]	● FAIL
Test 2b fuzzy positives (the result)	20/37	0.557	[0.41, 0.70]	0.793	-0.236	[-0.38, -0.08]	● FAIL
Test 3 vs gold-standard negatives	29/12	0.310	[0.15, 0.49]	0.569	-0.259	[-0.53, +0.03]	● FAIL
Test 4 vs scrambles (separate arm)	29/9	0.667	[0.44, 0.87]	--	--	--	
Homotypic arm	14/14	0.543	[0.31, 0.77]	0.883	-0.339	[-0.58, -0.11]	● FAIL
Heterotypic arm	15/29	0.607	[0.43, 0.77]	0.556	+0.051	[-0.17, +0.28]	● FAIL

Test 4 has no baseline column by design: scrambles are composition-matched, so the composition model cannot separate them from their parents and is not a meaningful comparator in that arm.

READS AS

The plan's own decision table says: delta-AUC_fuzzy near zero with AUC_fold high is the clean expected split; both low means debug first. Here delta-AUC_fuzzy is -0.11 with a CI crossing zero, and AUC_fold is 0.42 - not high. Pages 3 to 4 show why that is not a broken pipeline but a real failure mode: AF3 dimerises any zipper-competent pair, cognate or not.

Does AF3 beat composition? Every grouping.

Delta-AUC = AUC(AF3 ipTM) - AUC(composition baseline), paired bootstrap, 5,000 resamples. Right of the zero line, AF3 adds information over composition; left, it subtracts. A group passes only if the whole interval sits right of zero.

Pre-registered

Test 1 17 positives vs 31 real negatives (29/43)

Test 2a mutual folding vs hard zipper (4/6)

Test 2b fuzzy positives (the result) (20/37)

Test 3 vs gold-standard negatives (29/12)

By pair type

Homotypic arm (14/14)

Heterotypic arm (15/29)

By negative type

vs noncognate (29/12)

vs cross_kingdom (29/5)

vs charge_repulsive (29/2)

vs hard_coiledcoil (29/6)

vs homo_negative (29/14)

By negative evidence

vs demonstrated (29/5)

vs demonstrated+matched_internal (29/7)

vs gold (both demonstrated grades) (29/12)

vs inferred (29/11)

vs absence_only (29/12)

vs uncertain (29/2)

vs weak (inferred + absence_only) (29/23)

By positive subgroup

Fuzzy heterotypic positives (15/43)

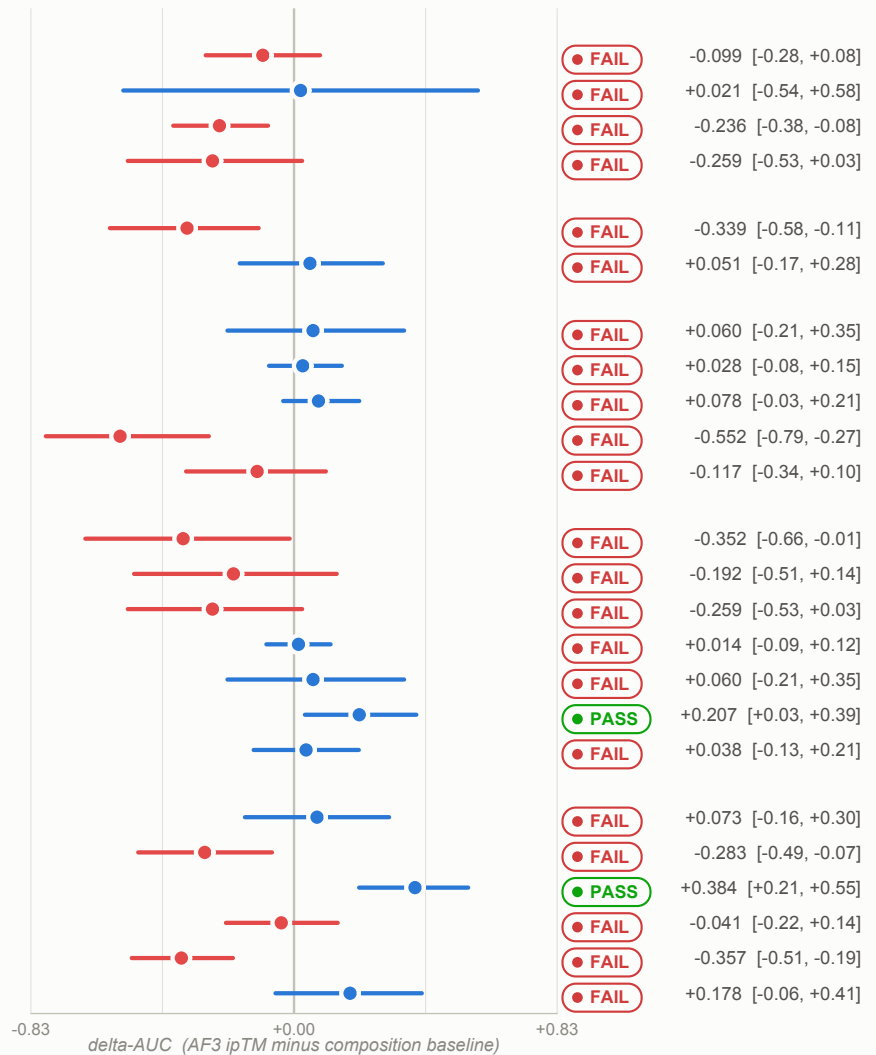
Fuzzy homotypic positives (14/43)

Mutual-folding (memorised) positives (4/43)

All 21 positives incl. memorised (33/43)

LLPS-reported positives (15/43)

Positives without LLPS report (14/43)



READS AS

Exactly one grouping clears zero on ipTM, and it is the pipeline control: the four memorised complexes, at +0.52. Of the other 23, two exclude zero on the wrong side - the homotypic arm (-0.34) and the gold-standard negatives (-0.52), where AF3 ranks the negatives above the positives - and the rest are indistinguishable from composition.

Is the pipeline sound?

Before reading a null result, the controls have to hold. They do: the canary folds, the memorised complexes score highest in the set, and every job ran the sequence it was supposed to.

Memorised controls and the canary

pipeline controls, excluded from every headline number

control	PDB	residues	ipTM	mean pLDDT	interchain PAE	ranking	sample sd
P21 GCN4_LZ x GCN4_LZ	2ZTA	66	0.73	88.1	4.1	1.24	0.027
P20 MAX_bHLHZ x MAX_bHLHZ	1AN2	162	0.80	86.8	6.1	0.95	0.005
P07 FOS_bZIP x JUN_bZIP	1FOS	129	0.74	86.6	8.5	1.03	0.005
P06 NCOA3_ACTR x CBP_NCBD	1KBH	106	0.70	76.9	6.3	0.89	0.021

GCN4-p1 (P21, 66 residues) is the pre-registered canary: if AF3 could not dimerise it confidently the instruction was to stop and debug. It reaches ipTM 0.73 with mean pLDDT 88 and a 4.1 Å interchain PAE, on an MSA of only 23 paired sequences. All four memorised complexes land between 0.70 and 0.80, the top of the whole set.

Data integrity

checked against the job requests the server returned

check	result	
Submitted sequences match the pair table	all 60 jobs, both chains exact	● PASS
Samples per job	5 of 5 for every job (300 summary files, 300 full-data files)	● PASS
Job-to-row matching	60 matched by name, 0 ambiguous	● PASS
Model seeds	59 jobs on seed 1; N24 on a server-assigned seed	● WARN
Steric clashes	a clash in some sample of P13 and P17 only (2 of 60)	● WARN
Structure templates	on for all 95 chain entries - server default, not a choice	● WARN
Jobs outstanding	S20 and S21 (batch 4) not yet run	● NOTE

Sample spread

five samples from one seed, so this is sample spread, not seed replication

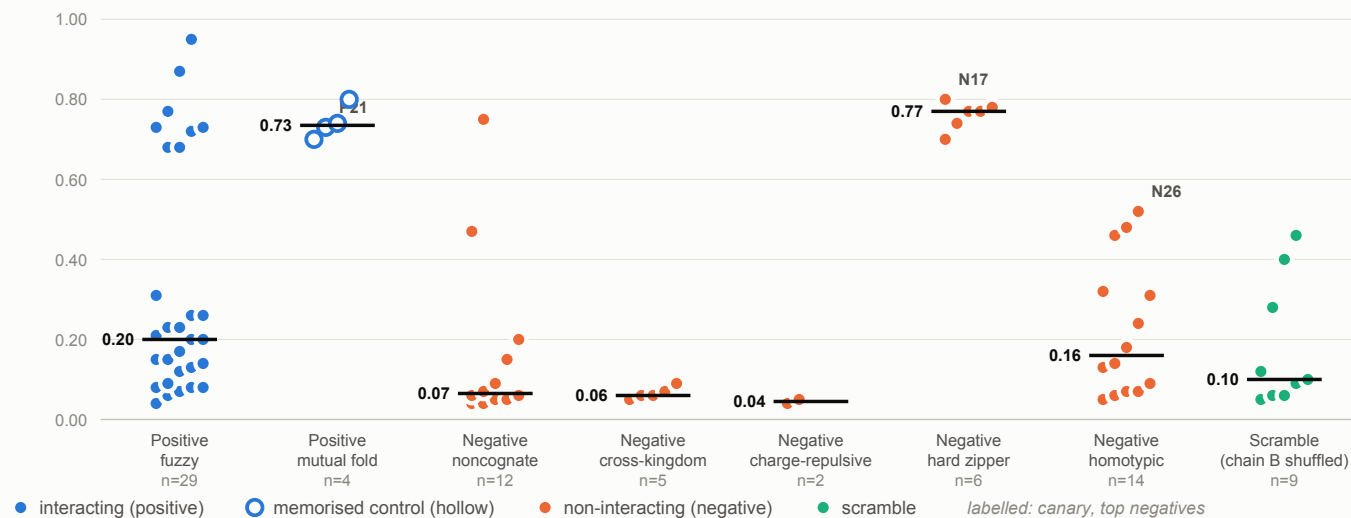
Fuzzy positives: mean within-job ipTM sd 0.0123 (n=20). Mutual-folding positives: 0.0147 (n=8). The plan expected fuzzy pairs to be the more variable; they are not, because all five samples come from one seed and share a trunk embedding. Seed replicates are still needed to measure run-to-run variance.

READS AS

Nothing here explains the null result away. The controls behave, so the failures on pages 1 and 2 are about AF3's discrimination, not about the harness - with one caveat that cuts the other way: templates were on for every chain, which can only help the controls look good.

Where every job actually lands

One dot per job, ipTM of the top-ranked sample. Black tick is the class median. The dots are the whole dataset: no aggregation, nothing hidden.



READS AS The three hard-zipper negatives (N15, N16, N17) sit at 0.70-0.80 - level with the memorised true complexes - while the 17 genuine fuzzy positives have a median ipTM of 0.20 (range 0.04-0.95). AF3 confidence tracks "can these two helices form a coiled coil", not "do these two regions interact".

Primary metric, full statistics *permutation p, BH-corrected across the 16-metric panel*

group	n	ipTM AUC	95% CI	perm p	BH q	composition	length alone
Test 1 17 positives vs 31 real negatives	29/43	0.593	[0.46, 0.72]	0.1761	0.3130	0.692	0.646
Test 2a mutual folding vs hard zipper	4/6	0.354	[0.00, 0.75]	0.5093	0.8920	0.333	0.417
Test 2b fuzzy positives (the result)	20/37	0.557	[0.41, 0.70]	0.4915	0.6455	0.793	0.739
Test 3 vs gold-standard negatives	29/12	0.310	[0.15, 0.49]	0.0628	0.1435	0.569	0.784
Test 4 vs scrambles (separate arm)	29/9	0.667	[0.44, 0.87]	0.1475	0.2622	--	0.626
Homotypic arm	14/14	0.543	[0.31, 0.77]	0.7038	0.8044	0.883	0.760
Heterotypic arm	15/29	0.607	[0.43, 0.77]	0.2483	0.4414	0.556	0.568

The last two columns are the reference predictors on the same jobs: a composition-only model, and total chain length on its own. Where they beat the ipTM column, AF3 has added nothing a sequence length could not.

Homotypic and heterotypic, analysed apart

Plan criterion 3. The two arms have different baselines and different MSA situations, so pooling them lets chain identity leak into the answer. Each arm is scored against a baseline refitted inside that arm.

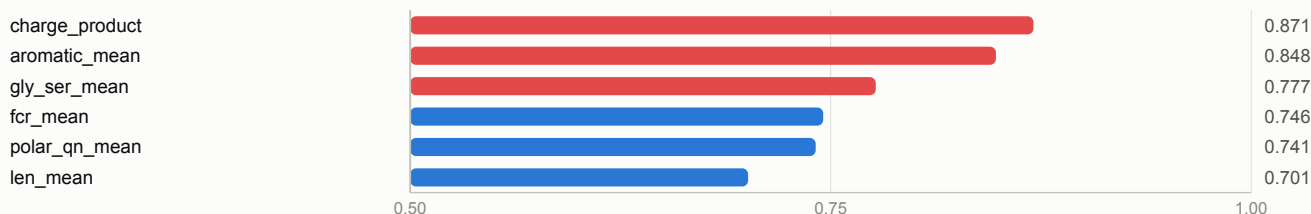
arm	n	ipTM AUC	95% CI	arm baseline	delta	delta 95% CI	verdict	contact sum AUC
Homotypic arm	14/14	0.543	[0.31, 0.77]	0.883	-0.339	[-0.58, -0.11]	FAIL	0.923
Heterotypic arm	15/29	0.607	[0.43, 0.77]	0.556	+0.051	[-0.17, +0.28]	FAIL	0.649

READS AS The homotypic arm is where the benchmark is decided, and it is where AF3 loses outright: ipTM 0.56 against a composition baseline of 0.90, delta -0.34 with the interval clear of zero. In the heterotypic arm composition is useless (0.50) and AF3 is no better (0.49) - on only 5 positives.

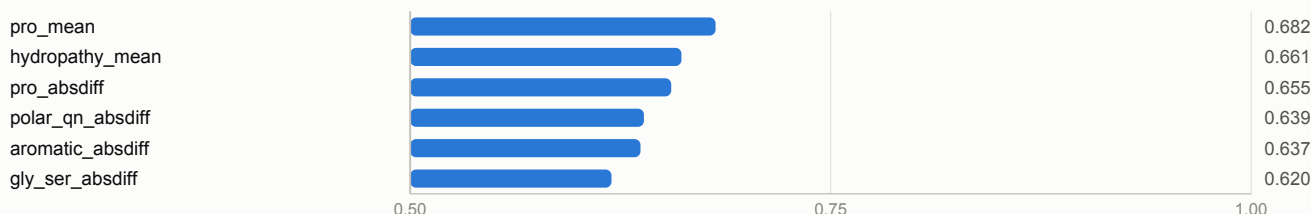
Why composition is so strong in the homotypic arm

single-feature AUC, two-sided, from baseline_leakage.py

Homotypic pairs (14 positive / 11 negative)



Heterotypic pairs (7 positive / 20 negative)



A single feature above 0.75, or a leave-one-out model above 0.80, is the documented leakage threshold. In the homotypic arm five features clear it on their own: LC domains that self-associate really are aromatic-rich and weakly charged. That is biology, not a curation error - but it means a homotypic claim needs to beat 0.85, not 0.5.

By type of negative

The plan asks for negative subtypes separately, because they are not equally clean: non-cognate human pairs are the fair test, cross-kingdom pairs share the scramble arm's missing-MSA problem in milder form, and the hard zippers are adversarial by design.

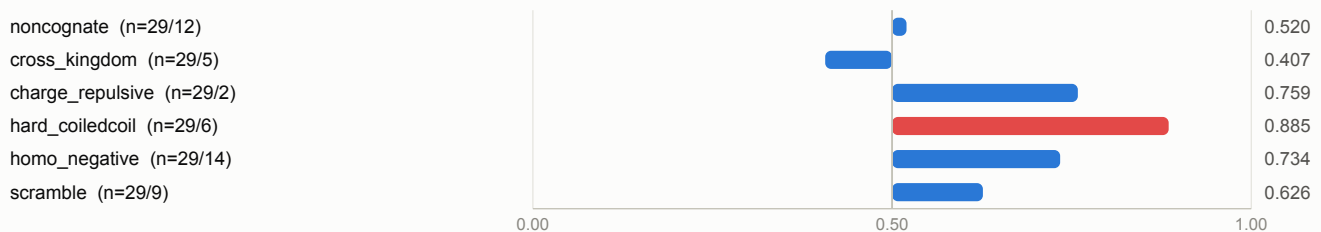
group	n	ipTM AUC	95% CI	BH q	compos.	delta	delta CI	length	contact s..
noncognate	29/12	0.750	[0.55, 0.92]	0.0264	0.690	+0.060	[-0.21, +0.35]	0.520	0.874
cross_kingdom	29/5	0.910	[0.79, 1.00]	0.0048	0.883	+0.028	[-0.08, +0.15]	0.407	0.934
charge_repulsive	29/2	0.974	[0.90, 1.00]	n<3	0.897	+0.078	[-0.03, +0.21]	0.759	0.983
hard_coiledcoil	29/6	0.103	[0.01, 0.22]	0.0043	0.655	-0.552	[-0.79, -0.27]	0.885	0.586
homo_negative	29/14	0.587	[0.40, 0.76]	0.5285	0.704	-0.117	[-0.34, +0.10]	0.734	0.635
scramble	29/9	0.667	[0.44, 0.87]	0.2622	--	--	--	0.626	0.745

Charge-repulsive has only 2 negatives, so no permutation test is reported for it. Scrambles have no composition comparator by construction.

READS AS The subtype decides the answer. AF3 scores highest against the negatives whose second chain has the weakest evolutionary signal - charge-repulsive (0.97, only 2 pairs) and cross-kingdom (0.91) - holds up against non-cognate human pairs (0.75), then inverts completely against the hard zippers (0.10, where length alone scores a perfect 0.89 because every zipper pair is short) and against the homotypic negatives (0.59). The cleanest negatives and the nastiest ones disagree, which is the whole reason the subtypes are reported apart.

The same groups, by what the model could be reading instead

Total chain length alone, as a classifier (0.5 = chance)



Bars to the right of the centre line mean the positives are longer; to the left, the negatives are. Two of these groups are separated almost perfectly by length before AF3 is consulted at all.

By how well we know the negative is negative

Every negative carries an evidence grade. 21 of 31 rest on inferred or absence-only evidence; 8 are demonstrated. The plan says that if the headline and the gold-standard subset disagree, believe the gold standard.

group	n	ipTM AUC	95% CI	BH q	compos.	delta	delta CI	length	contact s..
demonstrated	29/5	0.345	[0.08, 0.63]	0.5807	0.697	-0.352	[-0.66, -0.01]	0.855	0.559
demonstrated+matched_internal	29/7	0.286	[0.13, 0.46]	0.2269	0.478	-0.192	[-0.51, +0.14]	0.734	0.685
gold (both demonstrated grades)	29/12	0.310	[0.15, 0.49]	0.1435	0.569	-0.259	[-0.53, +0.03]	0.784	0.632
inferred	29/11	0.882	[0.75, 0.99]	0.0006	0.868	+0.014	[-0.09, +0.12]	0.567	0.839
absence_only	29/12	0.750	[0.55, 0.92]	0.0264	0.690	+0.060	[-0.21, +0.35]	0.520	0.874
uncertain	29/2	0.845	[0.67, 0.98]	n<3	0.638	+0.207	[+0.03, +0.39]	0.259	0.741
weak (inferred + absence_only)	29/23	0.813	[0.68, 0.93]	0.0002	0.775	+0.038	[-0.13, +0.21]	0.543	0.857

Uncertain has 2 rows (H1.0 alone, ERD10 alone) and is descriptive only.

READS AS They disagree violently - and the gold standard is the confounded one. Against the 8 demonstrated negatives ipTM inverts to 0.12, but those 8 negatives are 94-236 residues while the 17 positives are 205-882, so length alone scores 0.985 on that exact split. The inversion is a length and foldability artefact, not evidence that AF3 ranks real complexes below non-complexes.

What the gold-standard subset is made of *the 8 demonstrated negatives, ranked by ipTM*

gold-standard negative	subtype	evidence	residues	ipTM	PAE	pLDDT	contact s..
N17 GCN4_LZ x JUN_bZIP	hard_coiledcoil	dem.	100	0.80	5.9	88	47
N15 FOS_bZIP x MAX_bHLHZ	hard_coiledcoil	dem.	143	0.78	6.1	88	73
N33 ESR1_LBD x NCOA2_NRbox2_LXX..	motif_dead_mutant	dem.+matched_in..	257	0.76	9.8	91	25
N16 JUN_bZIP x MAX_bHLHZ	hard_coiledcoil	dem.	148	0.70	9.4	87	72
N35 YWHAZ_1433 x RAF1_S259A	motif_dead_mutant	dem.+matched_in..	260	0.70	12.0	88	44
N36 CDC20_WD40 x BUB1B_KENAAA	motif_dead_mutant	dem.+matched_in..	452	0.55	20.2	76	55
N26 NCOA3_ACTR x NCOA3_ACTR	homo_negative	dem.+matched_in..	94	0.52	8.5	70	62
N27 CBP_NCBD x CBP_NCBD	homo_negative	dem.+matched_in..	118	0.46	12.5	67	52
N28 FOS_bZIP x FOS_bZIP	homo_negative	dem.+matched_in..	124	0.31	20.7	74	61
N32 MDM2_Ndomain x P53_TAD_L22Q..	motif_dead_mutant	dem.+matched_in..	170	0.17	24.8	66	53
N31 EIF4EBP1 x EIF4EBP1	homo_negative	dem.	236	0.14	27.1	48	111
N29 P53_TAD x P53_TAD	homo_negative	dem.	122	0.13	22.2	55	64

Six of the eight are short helix- or zipper-competent constructs. Three of them (N15, N16, N17) are exactly the adversarial coiled-coil negatives, and three more (N26, N27, N28) are single chains of a real complex run against themselves.

By type of positive

All rows below are scored against the same 31 real-sequence negatives, so the groups are directly comparable.

group	n	ipTM AUC	95% CI	BH q	compos.	delta	delta CI	length	contact s..
Fuzzy heterotypic positives	15/43	0.614	[0.45, 0.77]	0.3652	0.541	+0.073	[-0.16, +0.30]	0.614	0.557
Fuzzy homotypic positives	14/43	0.571	[0.42, 0.71]	0.4609	0.854	-0.283	[-0.49, -0.07]	0.681	0.962
Mutual-folding (memorised) posi..	4/43	0.872	[0.75, 0.97]	0.0147	0.488	+0.384	[+0.21, +0.55]	0.180	0.808
All 21 positives incl. memorised	33/43	0.627	[0.50, 0.75]	0.1079	0.667	-0.041	[-0.22, +0.14]	0.590	0.759
LLPS-reported positives	15/43	0.446	[0.31, 0.59]	0.7817	0.803	-0.357	[-0.51, -0.19]	0.746	0.782
Positives without LLPS report	14/43	0.751	[0.61, 0.87]	0.0105	0.573	+0.178	[-0.06, +0.41]	0.540	0.721

The LLPS rows are a stratification, never an outcome: a 1:1 co-fold cannot report on phase separation, which is many-body and concentration-dependent.

READS AS The four memorised complexes are the positives AF3 separates cleanly (ipTM AUC 0.87); fuzzy positives sit at 0.61 heterotypic and 0.57 homotypic, and reported LLPS makes no difference (0.45). Memorisation, not disorder, is what ipTM tracks here. One exception is worth flagging honestly: for the 12 fuzzy homotypic positives, contact probability sum beats composition by +0.108 [+0.018, +0.216] - the only cell outside the memorised controls to clear zero in all 464. It is exploratory, uncorrected for the 464 comparisons, and the pre-registered homotypic arm (which uses homotypic negatives and the arm baseline) does not pass. Treat it as the hypothesis to test on a fresh set, not a result.

The sharpest tests in the set *matched internal contrasts - same chains, same MSAs, only the pairing differs*

These three comparisons do not rest on a literature absence at all. ACTR and NCBD each fold only with the other, and c-Fos forms no stable homodimer, so ACTR x ACTR, NCBD x NCBD and Fos x Fos are negatives whose chains also appear in a positive. The plan calls the outcome unambiguous either way.

job	label	residues	ipTM	interchain PAE	mean pLDDT	contact sum
P06 NCOA3_ACTR x CBP_NCBD - ACTR and NCBD each fold only with the other						
P06 NCOA3_ACTR x CBP_NCBD	positive	106	0.70	6.3	76.9	66
N26 NCOA3_ACTR x NCOA3_ACTR	negative	94	0.52	8.5	70.3	62
N27 CBP_NCBD x CBP_NCBD	negative	118	0.46	12.5	67.3	52
P07 FOS_bZIP x JUN_bZIP - c-Fos forms no stable homodimer (O'Shea 1989)						
P07 FOS_bZIP x JUN_bZIP	positive	129	0.74	8.5	86.6	66
N28 FOS_bZIP x FOS_bZIP	negative	124	0.31	20.7	74.4	61

READS AS AF3 gets all three orderings right on ipTM, and on 12 of 16 metrics for P06 and 15 of 16 for P07. This is the one genuinely encouraging result in stage 1 - with the caveat on page 11 that both positives were handed per-chain templates taken from their own cognate complex.

The whole 16-metric panel

ipTM may simply be floored for disordered pairs, so the plan pre-registered a 16-metric panel with permutation p-values and Benjamini-Hochberg correction. Every metric is oriented so that a higher AUC means "more interaction".

AUC by metric and group

A dot in the corner marks BH $q < 0.05$ within that group. Cell values are AUCs; .50 is chance.

	ipTM	chain-pair ipTM	ranking score	PTM	fraction disordered	chain-pair PAE min	interchain PAE mean	interchain PAE p05	PAE frac < 10 A	PAE frac < 5 A	contact prob. sum	contact prob. density	contact prob. max	contacts > 0.5	contact particip. ratio	mean pLDDT
Test 1 (17/31)	.59	.59	.58	.53	.46	.71*	.53	.59	.61	.64	.75*	.55	.69*	.69*	.61	.43
Test 2a (4/3)	.35	.35	.48	.40	.54	.65	.62	.62	.67	.79	.42	.75	.52	.46	.71	.21
Test 2b (17/28)	.56	.56	.55	.44	.36	.70	.45	.52	.57	.62	.80*	.52	.69	.71*	.69	.29
Test 3 (17/8)	.31	.31	.38	.26	.33	.51	.24	.30	.37	.43	.63	.29	.47	.51	.59	.26
Test 4 (17/8)	.67	.67	.75	.63	.45	.82*	.59	.65	.67	.77*	.75	.55	.81*	.78*	.55	.65
Homotypic arm (12/11)	.54	.54	.63	.39	.30	.81*	.44	.57	.70	.68	.92*	.47	.78*	.79*	.54	.31
Heterotypic arm (5/20)	.61	.61	.55	.62	.58	.68	.57	.59	.58	.62	.65	.57	.66	.61	.62	.55

AUC, diverging around chance (0.50)
 0.15 inverted discriminating 0.85

Delta-AUC against the composition baseline, same cells

Blank row for test 4: scrambles have no composition comparator.

	ipTM	chain-pair ipTM	ranking score	PTM	fraction disordered	chain-pair PAE min	interchain PAE mean	interchain PAE p05	PAE frac < 10 A	PAE frac < 5 A	contact prob. sum	contact prob. density	contact prob. max	contacts > 0.5	contact particip. ratio	mean pLDDT
Test 1 (17/31)	-.10	-.10	-.11	-.17	-.23	.01	-.16	-.11	-.08	-.06	.06	-.14	-.00	-.00	-.08	-.27
Test 2a (4/3)	.02	.02	.15	.06	.21	.31	.29	.29	.33	.46	.08	.42	.19	.12	.38	-.12
Test 2b (17/28)	-.24	-.24	-.24	-.35	-.43	-.09	-.34	-.28	-.22	-.18	.01	-.28	-.10	-.09	-.10	-.50
Test 3 (17/8)	-.26	-.26	-.19	-.31	-.24	-.06	-.33	-.27	-.20	-.14	.06	-.28	-.10	-.06	.03	-.30
Test 4 (17/8)																
Homotypic arm (12/11)	-.34	-.34	-.25	-.49	-.58	-.08	-.44	-.32	-.18	-.20	.04	-.41	-.11	-.10	-.35	-.57
Heterotypic arm (5/20)	.05	.05	-.00	.06	.03	.13	.01	.04	.02	.06	.09	.01	.10	.05	.06	-.00

delta-AUC
 -0.30 worse than composition better +0.30

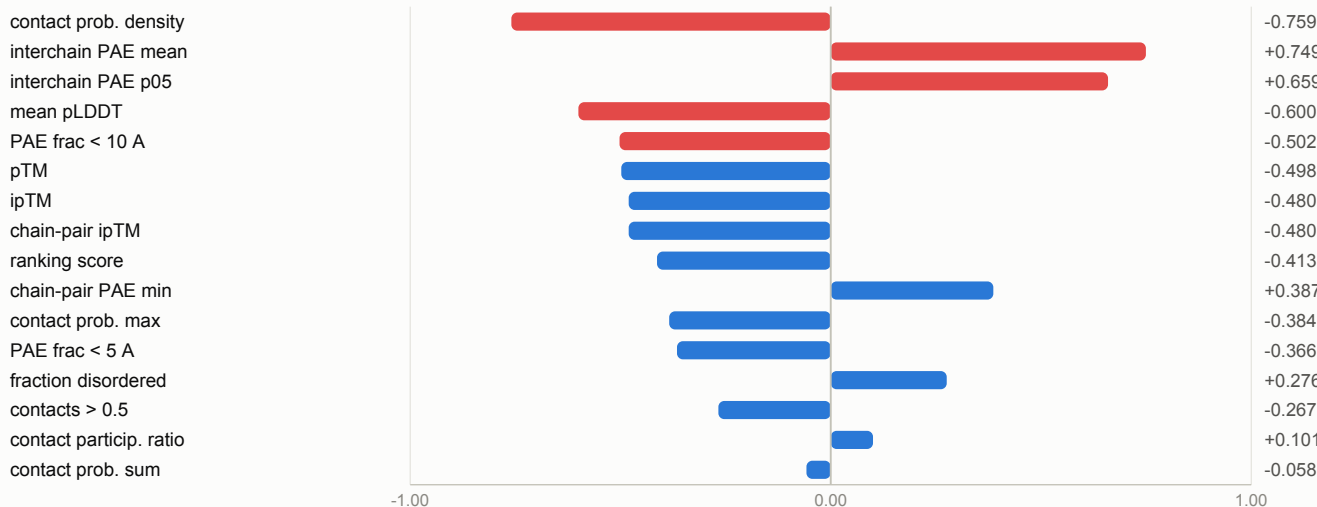
READS AS

Read the two grids together. Several metrics beat chance in the top grid - contact probability sum reaches 0.79 at $q = 0.008$, and 0.97 in the homotypic arm - yet the bottom grid is essentially all grey or red: across these seven groups not one metric beats composition with a CI clear of zero. The best single cell is PAE frac < 5 A at delta +0.458 (test 2a, $n=4/3$), and its interval still crosses zero. The one non-control exception anywhere in the 464 cells is on page 8.

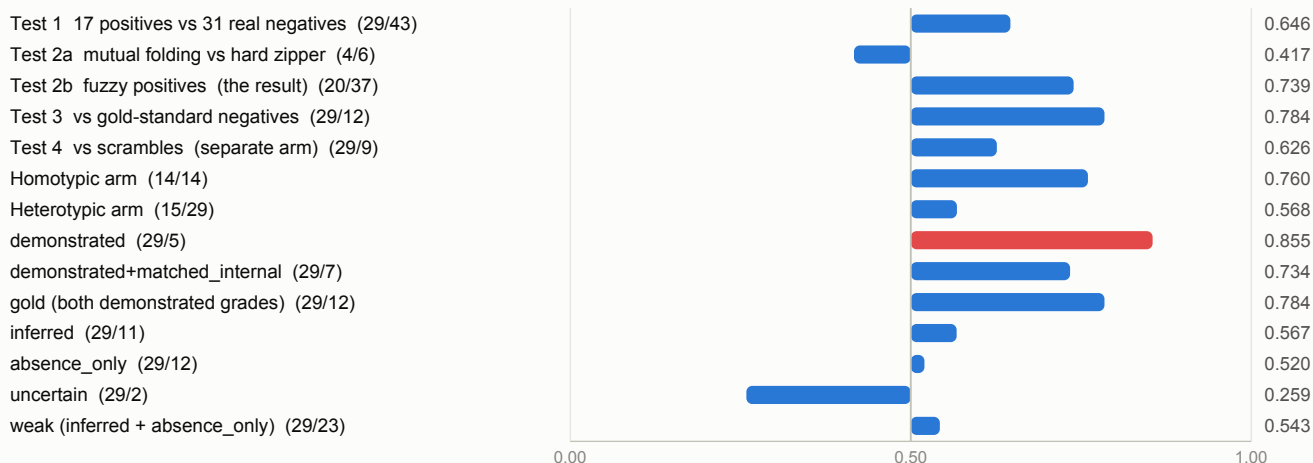
Confound 1: chain length

The plan flags any metric whose Spearman rho against total residue count exceeds 0.5, and requires it to be reported partialled as well as raw. Four metrics are flagged - and they are exactly the ones that carry the gold-standard inversion.

Spearman rho between metric and total residue count, all 60 jobs

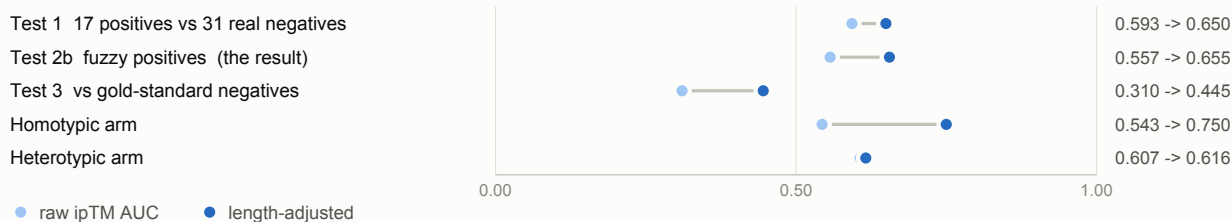


Length alone, as a classifier, in every group



Raw versus length-adjusted

metrics residualised on log total residues, then rescored



READS AS

Adjusting for length moves the gold-standard test from 0.12 to 0.45 and barely touches the headline test. So the dramatic inversion was length; the null result was not. This is a sensitivity analysis, not a pre-registered test.

Confound 2: what the model was given

Two inputs differ systematically between the arms of this benchmark, and neither was a design decision.

Structure templates were on for every chain server default; the batch files never set it

The uploaded batch files are dialect version 1 and contain no template switch. The server upgraded them to version 3 and set `useStructureTemplate = true` on all 95 chain entries, then returned 2 to 4 template hits per chain for all 60 jobs. For the matched internal negatives that matters directly: 6es7 - a structure containing both ACTR and NCBD, since it templates both chains of P06 - is also a template for N26 (ACTR x ACTR) and N27 (NCBD x NCBD), and 1fos, which templates both chains of P07, is a template in N15, N16, N17 and N28. AF3 applies templates within a chain, so this is the bound conformation of each partner rather than the interface itself - but it is exactly the help a zipper needs to dimerise.

job	label	template hits per chain (PDB entry)	unpaired MSA dept..	ipTM
P06 NCOA3_ACTR x CBP_NCBD	positive	a: 6es7, 6es5, 2c52, 3t47; b: 6es7, 6es6, 6sqc, 1jjs	a 379 b 774	0.70
N26 NCOA3_ACTR x NCOA3_ACTR	negative	a_b: 6es7, 6es5, 2c52, 3t47	a_b 379	0.52
N27 CBP_NCBD x CBP_NCBD	negative	b_a: 6es7, 6es6, 6sqc, 1jjs	b_a 774	0.46
P07 FOS_bZIP x JUN_bZIP	positive	a: 1fos, 5vpa, 1s9k, 1a02; b: 5vpa, 5t01, 1fos, 1t2k	a 1,478 b 1,259	0.74
N28 FOS_bZIP x FOS_bZIP	negative	b_a: 1fos, 5vpa, 1s9k, 1a02	b_a 1,478	0.31
N15 FOS_bZIP x MAX_bHLHZ	negative	a: 1fos, 5vpa, 1s9k, 1a02; b: 1r05, 1nlw, 1hlo, 5i4z	a 1,478 b 2,479	0.78
N17 GCN4_LZ x JUN_bZIP	negative	a: 5apu, 5apu, 5apw, 5apq; b: 5vpa, 5t01, 1fos, 1t2k	a 47 b 1,259	0.80
P21 GCN4_LZ x GCN4_LZ	positive	a_b: 5apu, 5apu, 5apw, 5apq	a_b 47	0.73

The scrambles really do lose their MSA confirming the plan's reason for keeping them a separate arm

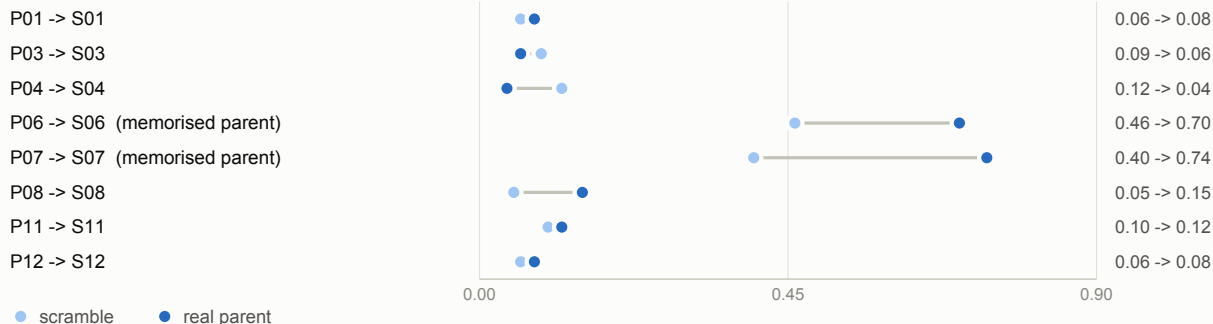
Unpaired MSA depth, shuffled chains: 1 to 34 sequences (median 8, n=9). Real chains: 1 to 16,479 (median 927, n=131). A shuffled sequence has no homologs, so low confidence there follows from missing evolutionary information rather than from a missing interface - which is why test 4 is reported apart and cannot carry the headline.

READS AS Both confounds push the same way: they make the memorised, foldable, short, template-rich pairs look good and the long fuzzy ones look bad. The single cheapest next experiment is to re-run a handful of jobs with `useStructureTemplate = false` and see how much of the control signal survives.

The scramble arm, paired

Each scramble shares chain A with its parent positive and differs only in a shuffled chain B, so the natural analysis is paired. Eight of the ten have run; S20 and S21 are pending.

ipTM: parent positive versus its composition-matched scramble



metric	parent beats scramble	sign-test p	median difference	unpaired AUC	BH q
ipTM	6 of 8	0.289	+0.020	0.667	0.2622
chain-pair ipTM	6 of 8	0.289	+0.020	0.667	0.2622
ranking score	6 of 8	0.289	+0.075	0.745	0.0728
pTM	6 of 8	0.289	+0.110	0.630	0.3386
fraction disordered	4 of 8	1.000	-0.010	0.446	0.6903
chain-pair PAE min	6 of 8	0.289	-5.910	0.816	0.0264
interchain PAE mean	6 of 8	0.289	-0.790	0.586	0.5608
interchain PAE p05	6 of 8	0.289	-4.050	0.648	0.2807
PAE frac < 10 A	4 of 8	1.000	+0.003	0.674	0.2622
PAE frac < 5 A	3 of 8	0.727	+0.000	0.766	0.0412
contact prob. sum	6 of 8	0.289	+9.985	0.745	0.0728
contact prob. density	6 of 8	0.289	+0.002	0.548	0.6903
contact prob. max	6 of 8	0.289	+0.315	0.814	0.0264
contacts > 0.5	4 of 8	1.000	+13.0	0.778	0.0412
contact particip. ratio	3 of 8	0.727	-248.0	0.550	0.6903
mean pLDDT	7 of 8	0.070	+7.822	0.648	0.2807

READS AS With 8 pairs the paired test has almost no power: ipTM favours the real parent in 6 of 8 (sign-test $p = 0.29$). There is no evidence either way yet on whether the little signal AF3 has depends on evolutionary information - finishing S20 and S21 makes it 10 pairs, which is still thin.

Local PAE: ipSAE and LIS

Added after the stage-1 numbers had been seen, so exploratory and outside the pre-registered panel. Both use only the confident interchain pairs - ipSAE those under 10 A (Dunbrack 2025), LIS those under 12 A (Kim et al. 2024) - so an unconfident flank cannot dilute a real interface the way it dilutes ipTM. Same top-ranked sample, same composition comparator.

The pre-registered tests, rescored delta-AUC against the same composition baseline

group	n	ipTM	ipSAE	delta	delta 95% CI		LIS	delta	delta 95% CI	
Test 1	29/43	0.593	0.682	-0.010	[-0.18, +0.16]	● FAIL	0.675	-0.017	[-0.19, +0.16]	● FAIL
Test 2a	4/6	0.354	0.667	+0.333	[-0.33, +0.92]	● FAIL	0.875	+0.542	[+0.17, +0.88]	● PASS
Test 2b	20/37	0.557	0.642	-0.151	[-0.29, -0.01]	● FAIL	0.645	-0.149	[-0.28, -0.01]	● FAIL
Test 3	29/12	0.310	0.497	-0.072	[-0.34, +0.20]	● FAIL	0.477	-0.092	[-0.37, +0.18]	● FAIL
Test 4	29/9	0.667	0.778	--	--		0.785	--	--	
Homotypic arm	14/14	0.543	0.724	-0.158	[-0.41, +0.10]	● FAIL	0.702	-0.181	[-0.44, +0.07]	● FAIL
Heterotypic arm	15/29	0.607	0.711	+0.155	[-0.06, +0.37]	● FAIL	0.672	+0.116	[-0.10, +0.34]	● FAIL

The ipTM column repeats page 1 for reference. Test 4 has no composition comparator. Every grouping, with q-values, is in appendix C.

Which PAE summary carries the signal AUC by group; rho is Spearman against total residues, all 60 jobs

metric	scores	Test 1	Test 2b	Homotypic arm	Heterotypic arm	rho vs length
ipTM	every pair	0.593	0.557	0.543	0.607	-0.48
interchain PAE mean	every pair	0.535	0.454	0.439	0.566	+0.75
PAE frac < 10 A	every pair	0.609	0.573	0.699	0.578	-0.50
chain-pair PAE min	single best pair	0.706	0.704	0.806	0.683	+0.39
ipSAE	pairs < 10 A	0.682	0.642	0.724	0.711	-0.27
LIS	pairs < 12 A	0.675	0.645	0.702	0.672	-0.40

Where the confident pairs are jobs with any interchain pair under each cutoff, by class

class	any pair < 10 A	median ipSAE	highest ipSAE	any pair < 12 A	median LIS
Positive, fuzzy heterotypic	10 of 15	0.531	0.850 P25	10 of 15	0.519
Positive, fuzzy homotypic	12 of 14	0.027	0.612 T03	13 of 14	0.260
Positive, memorised control	4 of 4	0.530	0.646 P20	4 of 4	0.621
Negative, noncognate	4 of 12	0.000	0.390 N37	4 of 12	0.000
Negative, cross-kingdom	0 of 5	0.000	--	0 of 5	0.000
Negative, charge-repulsive	0 of 2	0.000	--	0 of 2	0.000
Negative, hard zipper	6 of 6	0.504	0.599 N15	6 of 6	0.568
Negative, homotypic	7 of 14	0.005	0.158 N27	9 of 14	0.060
Scramble	3 of 9	0.000	0.119 S06	3 of 9	0.000

The sharpest contrasts matched internal negatives and the scramble arm

Both matched contrasts keep the right order on ipSAE and LIS. Fos x Jun sharpens: ipSAE 0.575 against Fos x Fos 0.012, where ipTM gave 0.74 against 0.31. ACTR x NCBD scores 0.486 against 0.141 and 0.158 for the two homodimers. In the scramble arm the parent wins 4 of 4 untied pairs (sign-test $p = 0.125$); the other 4 score exactly zero on both sides. The homotypic confident pairs vanish with the scramble (P08 0.030 $\rightarrow$ 0.000, P11 0.016 $\rightarrow$ 0.000), which still cannot tell sequence-specific self-pairing from an identical-chain effect.

READS AS

Scoring only the confident pairs changes no verdict. It lifts test 2b from 0.56 to 0.64 and the homotypic arm from 0.54 to 0.72 - enough that its delta interval now includes zero (-0.16 [-0.41, +0.10]) where ipTM's excluded it, not enough to beat composition - and drops the heterotypic arm from 0.61 to 0.71. Outside the memorised controls 5 cells beat composition. The class table is why: 5 of 15 heterotypic fuzzy positives have no interchain pair under 10 A, and the homotypic ones top out at ipSAE 0.612. A local score can only sharpen an interface AF3 has actually built.

How every number here was produced

Nothing in this report is computed by hand. The statistics come from the project's own functions so the report cannot drift from the pipeline.

Data

60 of 62 AlphaFold Server jobs, five samples each, downloaded 14 September 2026. Per job the reported value is the sample with the highest ranking_score, matching parse_af3_results.py. S20 and S21 (scrambles of the two memorised homodimers) are not yet run.

AUC

Mann-Whitney AUC with tie correction, on the metric oriented so higher means more interaction. 0.5 is chance; below 0.5 means the negatives outrank the positives.

Confidence intervals

Percentile bootstrap, 10,000 resamples, resampling positives and negatives independently.

p and q

Two-sided permutation test on the label assignment, 10,000 permutations, then Benjamini-Hochberg across the 16-metric panel within each group. Groups with fewer than 3 on either side are reported as descriptive only. ipSAE and LIS are corrected against the same 16 plus their own 2, so the panel q-values are untouched.

Composition baseline

Leave-one-out logistic regression on 24 cheap pair features (NCPR, FCR, aromatic and G/S content, hydropathy, length, charge complementarity, homotypic flag - means and absolute differences per pair). Fitted on the 52 real-sequence jobs for pooled groups, and refitted inside each arm for the homotypic and heterotypic rows. Raw LOO AUC: pooled 0.700, homotypic 0.851, heterotypic 0.471 (0.529 two-sided).

Delta-AUC

Paired bootstrap on AUC(AF3) - AUC(baseline), 5,000 resamples, resampling the same jobs for both score vectors. A test passes only if the 95% interval excludes zero. This is the pre-registered primary criterion; beating 0.5 is not the bar.

Local PAE scores

Both come from the full PAE matrix of the same top-ranked sample. ipSAE follows ipsae.py (Dunbrack 2025), d0res variant: for each residue, partners with PAE < 10 Å are kept, d0 is set from their number (floored at 26 residues and 1 Å), $1/(1+(PAE/d0)^2)$ is averaged over them, and the score is the best residue in the better direction. LIS follows Kim et al. 2024: the mean of $1 - PAE/12$ over interchain pairs with PAE below 12 Å, per direction, averaged over both. Both match a residue-by-residue transcription of the reference code on all 60 jobs to 1e-16.

Pre-registered versus exploratory

Pre-registered: the five subsets on page 1, the homotypic/heterotypic split, the 16-metric panel with BH correction, and the length confound flag. Exploratory, and labelled as such: the negative-subtype, evidence-grade and positive-subgroup groupings, the length-adjusted sensitivity analysis, the paired scramble test, and the two local PAE scores on page 13 and in appendix C, which were added after the stage-1 numbers had been seen.

Known caveats carried from the plan

Structure templates were on for all 95 chain entries (server default). 59 jobs ran on seed 1 and N24 on a server-assigned seed. Five samples share one seed, so spread is sample spread, not seed replication. 21 of 31 negatives rest on inferred or absence-only evidence. Length is not matched by design. Two negatives (H1.0 alone, ERD10 alone) are graded uncertain and should be confirmed or dropped. has_clash appears in the plan's metric list but not in the parser's panel; it is true in some sample of P13 and P17 only.

Files

af3_report_data.json holds every number in this report; af3_all_statistics.csv is the full table - 496 panel cells (31 groups x 16 metrics), then 62 exploratory rows with metric ipsae or lis; af3_scored.csv is the per-job scored table written by parse_af3_results.py, with ipsae and lis as its last columns.

Appendix A - every job, every headline metric

All 60 scored jobs. This is the table view for every figure in the report: each plotted dot appears here as a number.

ID	chains	label	class	res	ipTM	sd	pTM	PAE	pLDDT	C sum	C>.5	disor..	comp..
N01	FUS_LC x PTMA_full	negative	noncognate	274	0.04	0.000	0.09	31.6	47.4	0	0	1.00	-3.86
N02	HNRNPA1_LC x H10_full	negative	noncognate	329	0.06	0.004	0.27	31.2	56.5	4	0	0.79	-3.03
N03	TDP43_CTD x NPM1_acidic	negative	noncognate	269	0.05	0.000	0.13	31.0	49.1	0	0	0.25	-2.55
N04	EWSR1_LC x SNCA_full	negative	noncognate	405	0.05	0.004	0.12	31.6	53.5	0	0	1.00	-0.40
N05	TAF15_LC x MAX_bHLHZ	negative	noncognate	289	0.20	0.029	0.29	29.1	55.0	35	0	0.81	-2.25
N06	DDX4_NIDR x H10_full	negative	noncognate	430	0.07	0.004	0.23	31.5	48.3	1	0	0.84	+0.03
N07	CAPRIN1_CT x GCN4_LZ	negative	cross_kingdom	136	0.09	0.038	0.24	28.4	68.2	26	0	1.00	-0.88
N08	G3BP1_IDR1 x NPM1_acidic	negative	charge_repulsive	200	0.04	0.000	0.10	30.4	56.6	0	0	0.38	-2.19
N09	PTMA_full x NPM1_acidic	negative	charge_repulsive	232	0.05	0.005	0.10	30.8	50.4	0	0	0.54	-1.50
N10	FUS_LC x ERD10_full	negative	cross_kingdom	423	0.06	0.000	0.12	31.6	52.3	0	0	1.00	-2.81
N11	TDP43_CTD x ERD10_full	negative	cross_kingdom	408	0.07	0.004	0.12	31.4	51.8	0	0	1.00	-1.90
N12	NSP1_FG x H10_full	negative	noncognate	369	0.09	0.009	0.27	30.4	45.8	0	0	0.56	-1.56
N13	SRSF1_RS x PTMA_full	negative	noncognate	163	0.06	0.007	0.10	27.9	51.4	12	0	0.50	+4.17
N14	U170K_RS x NPM1_acidic	negative	noncognate	328	0.15	0.016	0.29	29.6	73.2	11	0	1.00	+0.15
N15	FOS_bZIP x MAX_bHLHZ	negative	hard_coiledcoil	143	0.78	0.008	0.78	6.1	88.5	73	70	0.38	-1.54
N16	JUN_bZIP x MAX_bHLHZ	negative	hard_coiledcoil	148	0.70	0.008	0.75	9.4	87.2	72	70	0.36	-0.35
N17	GCN4_LZ x JUN_bZIP	negative	hard_coiledcoil	100	0.80	0.007	0.74	5.9	87.5	47	45	0.76	+0.20
N18	HNRNPA2_LC x ERD10_full	negative	cross_kingdom	420	0.06	0.000	0.12	31.5	52.3	0	0	1.00	-3.39
N19	UBQLN2_CT x SNCA_full	negative	noncognate	315	0.04	0.004	0.18	31.6	62.7	16	0	0.30	+0.98
N20	CAPRIN1_CT x ERD10_full	negative	cross_kingdom	363	0.05	0.000	0.11	31.5	57.3	0	0	1.00	-1.27
N21	PTMA_full x PTMA_full	negative	homo_negative	222	0.06	0.006	0.13	30.8	53.2	37	0	1.00	-4.82
N22	H10_full x H10_full	negative	homo_negative	388	0.07	0.005	0.24	30.9	52.4	32	0	0.63	-0.94
N23	NPM1_acidic x NPM1_acidic	negative	homo_negative	242	0.07	0.005	0.10	29.9	51.5	39	0	0.96	-1.23
N24	ERD10_full x ERD10_full	negative	homo_negative	520	0.09	0.005	0.12	31.2	44.1	44	0	1.00	+0.24
N25	G3BP1_IDR1 x G3BP1_IDR1	negative	homo_negative	158	0.05	0.005	0.10	29.2	52.8	34	0	1.00	-5.54
N26	NCOA3_ACTR x NCOA3_ACTR	negative	homo_negative	94	0.52	0.136	0.55	8.5	70.3	62	1	0.81	-1.41
N27	CBP_NCBD x CBP_NCBD	negative	homo_negative	118	0.46	0.046	0.56	12.5	67.3	52	18	0.27	+0.87
N28	FOS_bZIP x FOS_bZIP	negative	homo_negative	124	0.31	0.006	0.51	20.7	74.4	61	53	0.66	-0.26
N29	P53_TAD x P53_TAD	negative	homo_negative	122	0.13	0.017	0.22	22.2	55.3	64	0	0.97	-2.85
N30	P21_full x P21_full	negative	homo_negative	328	0.24	0.042	0.27	27.6	46.6	148	2	0.81	+0.45
N31	EIF4EBP1 x EIF4EBP1	negative	homo_negative	236	0.14	0.019	0.24	27.1	48.2	111	0	0.66	+0.37
N32	MDM2_Ndomain x P53_TAD_L22QW..	negative	motif_dead_mutant	170	0.17	0.020	0.52	24.8	65.6	53	0	0.48	+2.16
N33	ESR1_LBD x NCOA2_NRbox2_LXXAA	negative	motif_dead_mutant	257	0.76	0.010	0.93	9.8	91.1	25	22	0.04	+0.11
N34	ABL1_SH3 x SH3BP1_PxxP_scr	scramble	scramble	71	0.28	0.015	0.73	12.9	87.1	26	1	0.14	--
N35	YWHAZ_1433 x RAF1_S259A	negative	motif_dead_mutant	260	0.70	0.009	0.88	12.0	88.3	44	32	0.11	+0.86
N36	CDC20_WD40 x BUB1B_KENAAA	negative	motif_dead_mutant	452	0.55	0.010	0.73	20.2	76.4	55	29	0.30	+2.57
N37	ABL1_SH3 x NCOA2_NRbox2	negative	noncognate	70	0.75	0.040	0.85	3.7	94.1	20	0	0.13	-2.00
N38	ESR1_LBD x BUB1B_KENbox	negative	noncognate	271	0.47	0.020	0.88	18.0	86.9	40	21	0.08	-0.42
P01	PTMA_full x H10_full	positive	fuzzy	305	0.08	0.004	0.29	30.8	51.5	3	0	0.76	-0.83
P02	SRSF1_RS x U170K_RS	positive	fuzzy	259	0.08	0.015	0.33	30.0	73.6	0	0	1.00	-2.05
P03	G3BP1_IDR1 x G3BP1_RGG	positive	fuzzy	205	0.06	0.007	0.36	30.1	62.6	13	0	0.64	-2.26
P04	FMR1_CT x CAPRIN1_CT	positive	fuzzy	291	0.04	0.006	0.09	31.4	50.1	0	0	0.25	-1.39
P05	SNCA_full x TAU_2N4R	positive	fuzzy	581	0.07	0.005	0.12	31.7	53.4	2	0	0.85	+0.27
P06	NCOA3_ACTR x CBP_NCBD	positive	mutual_folding	106	0.70	0.021	0.72	6.3	76.9	66	48	0.37	-0.98

PAE = mean interchain PAE (Å). C sum = interchain contact-probability sum. C>.5 = interchain token pairs above 0.5. compos. = composition-baseline score (leave-one-out, pooled fit). sd = spread across the five samples.

Appendix A continued

ID	chains	label	class	res	ipTM	sd	pTM	PAE	pLDDT	C sum	C>.5	disor..	comp..
P07	FOS_bZIP x JUN_bZIP	positive	mutual_folding	129	0.74	0.005	0.71	8.5	86.6	66	55	0.60	-1.12
P08	FUS_LC x FUS_LC	positive	fuzzy	326	0.15	0.007	0.16	27.5	49.5	321	46	1.00	+3.30
P09	HNRNPA1_LC x HNRNPA1_LC	positive	fuzzy	270	0.21	0.026	0.21	26.1	50.8	392	81	1.00	+3.05
P10	HNRNPA2_LC x HNRNPA2_LC	positive	fuzzy	320	0.15	0.019	0.15	28.2	47.1	385	61	1.00	+3.60
P11	TDP43_CTD x TDP43_CTD	positive	fuzzy	296	0.12	0.010	0.17	28.4	44.9	305	28	1.00	+2.65
P12	DDX4_NIDR x DDX4_NIDR	positive	fuzzy	472	0.08	0.000	0.09	31.0	43.5	101	0	1.00	+1.14
P13	EWSR1_LC x EWSR1_LC	positive	fuzzy	530	0.13	0.006	0.14	30.1	47.5	301	30	1.00	+2.99
P14	TAF15_LC x TAF15_LC	positive	fuzzy	416	0.14	0.011	0.14	29.1	40.6	268	25	1.00	+3.19
P15	UBQLN2_CT x UBQLN2_CT	positive	fuzzy	350	0.17	0.010	0.20	28.9	58.3	203	23	0.59	+0.89
P16	NSP1_FG x NSP1_FG	positive	fuzzy	350	0.20	0.007	0.32	25.4	40.5	155	0	0.25	+2.19
P17	TAU_2N4R x TAU_2N4R	positive	fuzzy	882	0.23	0.034	0.23	30.3	44.1	291	84	0.96	+2.06
P18	SNCA_full x SNCA_full	positive	fuzzy	280	0.23	0.053	0.23	27.3	55.8	268	165	1.00	-0.71
P19	CAPRIN1_CT x CAPRIN1_CT	positive	fuzzy	206	0.09	0.017	0.11	28.2	49.2	92	0	1.00	+0.72
P20	MAX_bHLHZ x MAX_bHLHZ	positive	mutual_folding	162	0.80	0.005	0.80	6.1	86.8	90	89	0.31	-0.61
P21	GCN4_LZ x GCN4_LZ	positive	mutual_folding	66	0.73	0.027	0.76	4.1	88.1	45	45	1.00	-0.63
P22	TP53_full x MDM2_full	positive	fuzzy	905	0.26	0.000	0.31	30.7	54.1	97	26	0.63	+1.19
P23	TP53_full x MDM2_full	positive	fuzzy	884	0.31	0.006	0.35	29.3	58.9	124	27	0.58	+2.42
P24	P53_TAD x MDM2_Ndomain	positive	coupled_folding	170	0.20	0.023	0.54	22.2	64.4	69	4	0.46	+0.64
P25	ESR1_LBD x NCOA2_NRbox2	positive	coupled_folding	257	0.95	0.005	0.94	4.2	93.5	24	24	0.04	-2.01
P26	ABL1_SH3 x SH3BP1_PxxP	positive	coupled_folding	71	0.87	0.004	0.89	2.3	95.9	30	28	0.14	-2.47
P27	YWHAZ_1433 x RAF1_pS259	positive	coupled_folding	260	0.73	0.060	0.88	11.5	88.4	45	32	0.11	-1.10
P28	CDC20_WD40 x BUB1B_KENbox	positive	coupled_folding	452	0.68	0.006	0.74	19.0	77.2	37	26	0.30	-0.52
S01	PTMA_full x H10_full_scr	scramble	scramble	305	0.06	0.004	0.15	31.3	56.8	0	0	1.00	--
S03	G3BP1_IDR1 x G3BP1_RGG_scr	scramble	scramble	205	0.09	0.009	0.19	26.6	39.1	76	0	0.44	--
S04	FMR1_CT x CAPRIN1_CT_scr	scramble	scramble	291	0.12	0.016	0.15	29.4	43.2	83	0	0.42	--
S06	NCOA3_ACTR x CBP_NCBD_scr	scramble	scramble	106	0.46	0.068	0.44	17.5	64.1	52	2	0.78	--
S07	FOS_bZIP x JUN_bZIP_scr	scramble	scramble	129	0.40	0.008	0.57	16.7	77.9	60	29	0.62	--
S08	FUS_LC x FUS_LC_scr	scramble	scramble	326	0.05	0.005	0.08	31.4	39.5	0	0	0.75	--
S11	TDP43_CTD x TDP43_CTD_scr	scramble	scramble	296	0.10	0.008	0.15	29.3	44.6	51	0	0.45	--
S12	DDX4_NIDR x DDX4_NIDR_scr	scramble	scramble	472	0.06	0.000	0.13	31.6	41.7	0	0	1.00	--
T01	NCOA3_ACTR x CBP_NCBD	positive	mutual_folding	106	0.73	0.027	0.74	5.8	77.8	66	48	0.38	-0.98
T02	FOS_bZIP x JUN_bZIP	positive	mutual_folding	129	0.72	0.008	0.70	9.1	86.6	67	55	0.59	-1.12
T03	MAX_bHLHZ x MAX_bHLHZ	positive	mutual_folding	162	0.77	0.004	0.78	6.6	85.2	91	89	0.31	-0.61
T04	GCN4_LZ x GCN4_LZ	positive	mutual_folding	66	0.68	0.020	0.72	4.8	86.4	46	41	1.00	-0.63
T05	FOS_bZIP x MAX_bHLHZ	negative	hard_coiledcoil	143	0.77	0.000	0.78	6.7	88.8	72	70	0.38	-1.54
T06	JUN_bZIP x MAX_bHLHZ	negative	hard_coiledcoil	148	0.74	0.005	0.77	7.8	88.3	72	72	0.43	-0.35
T07	GCN4_LZ x JUN_bZIP	negative	hard_coiledcoil	100	0.77	0.008	0.72	6.5	86.4	48	45	0.76	+0.20
T08	NCOA3_ACTR x NCOA3_ACTR	negative	homo_negative	94	0.48	0.090	0.54	11.1	71.6	59	1	0.60	-1.41
T09	CBP_NCBD x CBP_NCBD	negative	homo_negative	118	0.18	0.034	0.31	19.0	55.2	87	0	0.47	+0.87
T10	TP53_full x MDM2_full	positive	fuzzy	905	0.26	0.000	0.31	30.9	54.5	86	27	0.69	+1.19
T11	FOS_bZIP x FOS_bZIP	negative	homo_negative	124	0.32	0.005	0.51	20.6	75.6	61	53	0.63	-0.26

PAE = mean interchain PAE (Å). C sum = interchain contact-probability sum. C>.5 = interchain token pairs above 0.5. compos. = composition-baseline score (leave-one-out, pooled fit). sd = spread across the five samples.

Appendix B - every group, every metric

31 groupings x 16 metrics = 496 cells, each with its own bootstrap interval, permutation p, BH-corrected q within the group, and delta against the composition baseline (464 cells have one). A PASS badge marks a delta interval clear of zero: 15 cells qualify, 14 of them in the memorised-control block and one outside it (fuzzy homotypic positives, contact probability sum). No pre-registered test passes.

Pre-registered / 1. positives vs real negatives (29 positive / 43 negative, length alone 0.646, composition 0.692)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	29/43	0.593	[0.46, 0.72]	0.1761	0.3130	-0.099	[-0.28, +0.08]
chain-pair ipTM	29/43	0.593	[0.46, 0.72]	0.1761	0.3130	-0.099	[-0.28, +0.08]
ranking score	29/43	0.582	[0.44, 0.71]	0.2346	0.3412	-0.110	[-0.30, +0.09]
pTM	29/43	0.526	[0.39, 0.66]	0.7021	0.7021	-0.166	[-0.37, +0.04]
fraction disordered	29/43	0.463	[0.33, 0.60]	0.5915	0.6570	-0.229	[-0.43, -0.02]
chain-pair PAE min	29/43	0.706	[0.58, 0.83]	0.0032	0.0203	+0.014	[-0.15, +0.19]
interchain PAE mean	29/43	0.535	[0.39, 0.67]	0.6159	0.6570	-0.157	[-0.36, +0.04]
interchain PAE p05	29/43	0.585	[0.45, 0.71]	0.2224	0.3412	-0.107	[-0.30, +0.09]
PAE frac < 10 A	29/43	0.609	[0.48, 0.73]	0.1053	0.2480	-0.083	[-0.27, +0.10]
PAE frac < 5 A	29/43	0.636	[0.51, 0.75]	0.0340	0.1088	-0.056	[-0.24, +0.13]
contact prob. sum	29/43	0.753	[0.63, 0.87]	0.0003	0.0048	+0.061	[-0.04, +0.17]
contact prob. density	29/43	0.554	[0.42, 0.68]	0.4398	0.5412	-0.138	[-0.32, +0.04]
contact prob. max	29/43	0.689	[0.56, 0.81]	0.0074	0.0296	-0.003	[-0.16, +0.16]
contacts > 0.5	29/43	0.692	[0.57, 0.81]	0.0038	0.0203	-0.000	[-0.16, +0.16]
contact particip. ratio	29/43	0.613	[0.48, 0.74]	0.1085	0.2480	-0.079	[-0.20, +0.05]
mean pLDDT	29/43	0.427	[0.29, 0.57]	0.2896	0.3861	-0.265	[-0.49, -0.04]

Pre-registered / 2a. mutual folding vs hard coiled-coil (4 positive / 6 negative, length alone 0.417, composition 0.333)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	4/6	0.354	[0.00, 0.75]	0.5093	0.8920	+0.021	[-0.54, +0.58]
chain-pair ipTM	4/6	0.354	[0.00, 0.75]	0.5093	0.8920	+0.021	[-0.54, +0.58]
ranking score	4/6	0.479	[0.08, 0.88]	0.9478	0.9478	+0.146	[-0.46, +0.67]
pTM	4/6	0.396	[0.00, 0.83]	0.6132	0.8920	+0.062	[-0.58, +0.67]
fraction disordered	4/6	0.542	[0.12, 0.96]	0.8788	0.9478	+0.208	[-0.42, +0.75]
chain-pair PAE min	4/6	0.646	[0.25, 0.96]	0.4746	0.8920	+0.313	[-0.42, +0.92]
interchain PAE mean	4/6	0.625	[0.21, 1.00]	0.6061	0.8920	+0.292	[-0.33, +0.83]
interchain PAE p05	4/6	0.625	[0.25, 0.92]	0.5533	0.8920	+0.292	[-0.37, +0.88]
PAE frac < 10 A	4/6	0.667	[0.25, 1.00]	0.4761	0.8920	+0.333	[-0.25, +0.83]
PAE frac < 5 A	4/6	0.792	[0.42, 1.00]	0.1725	0.8920	+0.458	[+0.00, +0.83]
contact prob. sum	4/6	0.417	[0.00, 0.83]	0.7639	0.9478	+0.083	[-0.54, +0.67]
contact prob. density	4/6	0.750	[0.38, 1.00]	0.2571	0.8920	+0.417	[-0.12, +0.88]
contact prob. max	4/6	0.521	[0.17, 0.92]	0.9271	0.9478	+0.188	[-0.46, +0.75]
contacts > 0.5	4/6	0.458	[0.00, 0.83]	0.8293	0.9478	+0.125	[-0.52, +0.71]
contact particip. ratio	4/6	0.708	[0.25, 1.00]	0.3010	0.8920	+0.375	[-0.17, +0.83]
mean pLDDT	4/6	0.208	[0.00, 0.54]	0.1728	0.8920	-0.125	[-0.67, +0.42]

Pre-registered / 2b. fuzzy positives vs real negatives (20 positive / 37 negative, length alone 0.739, composition 0.793)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	20/37	0.557	[0.41, 0.70]	0.4915	0.6455	-0.236	[-0.38, -0.08]
chain-pair ipTM	20/37	0.557	[0.41, 0.70]	0.4915	0.6455	-0.236	[-0.38, -0.08]
ranking score	20/37	0.551	[0.40, 0.71]	0.5244	0.6455	-0.242	[-0.41, -0.06]
pTM	20/37	0.445	[0.30, 0.60]	0.5055	0.6455	-0.349	[-0.53, -0.15]
fraction disordered	20/37	0.363	[0.23, 0.51]	0.0818	0.1636	-0.430	[-0.61, -0.24]
chain-pair PAE min	20/37	0.704	[0.55, 0.85]	0.0126	0.0504	-0.089	[-0.23, +0.05]
interchain PAE mean	20/37	0.454	[0.31, 0.61]	0.5746	0.6567	-0.339	[-0.50, -0.17]
interchain PAE p05	20/37	0.516	[0.37, 0.67]	0.8404	0.8404	-0.277	[-0.44, -0.11]
PAE frac < 10 A	20/37	0.573	[0.43, 0.72]	0.3278	0.5827	-0.220	[-0.36, -0.07]
PAE frac < 5 A	20/37	0.618	[0.49, 0.76]	0.0818	0.1636	-0.175	[-0.32, -0.02]
contact prob. sum	20/37	0.799	[0.64, 0.93]	<0.0001	0.0016	+0.006	[-0.10, +0.12]
contact prob. density	20/37	0.517	[0.36, 0.67]	0.8375	0.8404	-0.276	[-0.43, -0.12]
contact prob. max	20/37	0.691	[0.54, 0.83]	0.0192	0.0520	-0.103	[-0.22, +0.02]
contacts > 0.5	20/37	0.707	[0.57, 0.84]	0.0023	0.0184	-0.086	[-0.21, +0.05]
contact particip. ratio	20/37	0.689	[0.54, 0.82]	0.0195	0.0520	-0.105	[-0.25, +0.04]
mean pLDDT	20/37	0.293	[0.16, 0.44]	0.0098	0.0504	-0.500	[-0.70, -0.27]

Appendix B continued

Pre-registered / 3. positives vs gold negatives (29 positive / 12 negative, length alone 0.784, composition 0.569)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	29/12	0.310	[0.15, 0.49]	0.0628	0.1435	-0.259	[-0.53, +0.03]
chain-pair ipTM	29/12	0.310	[0.15, 0.49]	0.0628	0.1435	-0.259	[-0.53, +0.03]
ranking score	29/12	0.384	[0.21, 0.57]	0.2584	0.3758	-0.185	[-0.48, +0.12]
pTM	29/12	0.259	[0.12, 0.42]	0.0141	0.1024	-0.310	[-0.57, -0.04]
fraction disordered	29/12	0.326	[0.17, 0.50]	0.0789	0.1578	-0.243	[-0.48, +0.02]
chain-pair PAE min	29/12	0.511	[0.33, 0.69]	0.9216	0.9216	-0.057	[-0.34, +0.23]
interchain PAE mean	29/12	0.239	[0.10, 0.39]	0.0079	0.1024	-0.330	[-0.60, -0.04]
interchain PAE p05	29/12	0.299	[0.15, 0.47]	0.0459	0.1435	-0.270	[-0.54, +0.01]
PAE frac < 10 Å	29/12	0.374	[0.19, 0.57]	0.2062	0.3299	-0.195	[-0.48, +0.09]
PAE frac < 5 Å	29/12	0.434	[0.24, 0.63]	0.5138	0.6324	-0.135	[-0.43, +0.16]
contact prob. sum	29/12	0.632	[0.46, 0.80]	0.1982	0.3299	+0.063	[-0.11, +0.25]
contact prob. density	29/12	0.293	[0.15, 0.45]	0.0397	0.1435	-0.276	[-0.55, +0.01]
contact prob. max	29/12	0.467	[0.28, 0.65]	0.7524	0.8599	-0.102	[-0.37, +0.18]
contacts > 0.5	29/12	0.513	[0.32, 0.70]	0.8965	0.9216	-0.056	[-0.31, +0.20]
contact particip. ratio	29/12	0.595	[0.41, 0.77]	0.3621	0.4828	+0.026	[-0.20, +0.25]
mean pLDDT	29/12	0.264	[0.11, 0.43]	0.0192	0.1024	-0.305	[-0.57, -0.01]

Pre-registered / 4. positives vs scrambles (29 positive / 9 negative, length alone 0.626)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	29/9	0.667	[0.44, 0.87]	0.1475	0.2622	--	--
chain-pair ipTM	29/9	0.667	[0.44, 0.87]	0.1475	0.2622	--	--
ranking score	29/9	0.745	[0.53, 0.93]	0.0265	0.0728	--	--
pTM	29/9	0.630	[0.41, 0.83]	0.2540	0.3386	--	--
fraction disordered	29/9	0.446	[0.25, 0.65]	0.6386	0.6903	--	--
chain-pair PAE min	29/9	0.816	[0.67, 0.93]	0.0033	0.0264	--	--
interchain PAE mean	29/9	0.586	[0.36, 0.80]	0.4557	0.5608	--	--
interchain PAE p05	29/9	0.648	[0.43, 0.85]	0.1930	0.2807	--	--
PAE frac < 10 Å	29/9	0.674	[0.45, 0.87]	0.1184	0.2622	--	--
PAE frac < 5 Å	29/9	0.766	[0.61, 0.90]	0.0103	0.0412	--	--
contact prob. sum	29/9	0.745	[0.57, 0.89]	0.0273	0.0728	--	--
contact prob. density	29/9	0.548	[0.30, 0.78]	0.6903	0.6903	--	--
contact prob. max	29/9	0.814	[0.67, 0.93]	0.0029	0.0264	--	--
contacts > 0.5	29/9	0.778	[0.62, 0.90]	0.0095	0.0412	--	--
contact particip. ratio	29/9	0.550	[0.31, 0.79]	0.6738	0.6903	--	--
mean pLDDT	29/9	0.648	[0.40, 0.87]	0.1891	0.2807	--	--

By pair type / Homotypic only (14 positive / 14 negative, length alone 0.760, composition 0.883)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	14/14	0.543	[0.31, 0.77]	0.7038	0.8044	-0.339	[-0.58, -0.11]
chain-pair ipTM	14/14	0.543	[0.31, 0.77]	0.7038	0.8044	-0.339	[-0.58, -0.11]
ranking score	14/14	0.630	[0.42, 0.83]	0.2571	0.4570	-0.253	[-0.49, -0.04]
pTM	14/14	0.393	[0.17, 0.62]	0.3459	0.5534	-0.490	[-0.74, -0.23]
fraction disordered	14/14	0.298	[0.10, 0.51]	0.0580	0.1829	-0.584	[-0.82, -0.31]
chain-pair PAE min	14/14	0.806	[0.62, 0.94]	0.0054	0.0427	-0.077	[-0.29, +0.13]
interchain PAE mean	14/14	0.439	[0.21, 0.66]	0.5868	0.7825	-0.444	[-0.69, -0.19]
interchain PAE p05	14/14	0.566	[0.34, 0.78]	0.5639	0.7825	-0.316	[-0.56, -0.07]
PAE frac < 10 Å	14/14	0.699	[0.49, 0.89]	0.0686	0.1829	-0.184	[-0.43, +0.07]
PAE frac < 5 Å	14/14	0.679	[0.48, 0.86]	0.0858	0.1961	-0.204	[-0.45, +0.05]
contact prob. sum	14/14	0.923	[0.80, 1.00]	<0.0001	0.0016	+0.041	[-0.09, +0.19]
contact prob. density	14/14	0.469	[0.24, 0.69]	0.8057	0.8057	-0.413	[-0.66, -0.16]
contact prob. max	14/14	0.776	[0.58, 0.94]	0.0122	0.0488	-0.107	[-0.30, +0.07]
contacts > 0.5	14/14	0.786	[0.60, 0.95]	0.0080	0.0427	-0.097	[-0.29, +0.08]
contact particip. ratio	14/14	0.536	[0.31, 0.76]	0.7665	0.8057	-0.347	[-0.59, -0.11]
mean pLDDT	14/14	0.311	[0.11, 0.53]	0.0997	0.1994	-0.571	[-0.83, -0.27]

Appendix B continued

By pair type / Heterotypic only (15 positive / 29 negative, length alone 0.568, composition 0.556)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	15/29	0.607	[0.43, 0.77]	0.2483	0.4414	+0.051	[-0.17, +0.28]
chain-pair ipTM	15/29	0.607	[0.43, 0.77]	0.2483	0.4414	+0.051	[-0.17, +0.28]
ranking score	15/29	0.552	[0.37, 0.73]	0.5835	0.5867	-0.005	[-0.24, +0.25]
pTM	15/29	0.620	[0.44, 0.79]	0.1967	0.4414	+0.063	[-0.17, +0.30]
fraction disordered	15/29	0.582	[0.40, 0.75]	0.3920	0.5226	+0.025	[-0.22, +0.27]
chain-pair PAE min	15/29	0.683	[0.50, 0.86]	0.0501	0.4414	+0.126	[-0.10, +0.36]
interchain PAE mean	15/29	0.566	[0.38, 0.74]	0.4895	0.5594	+0.009	[-0.24, +0.26]
interchain PAE p05	15/29	0.594	[0.41, 0.77]	0.3065	0.4904	+0.038	[-0.22, +0.29]
PAE frac < 10 A	15/29	0.578	[0.40, 0.75]	0.3781	0.5226	+0.022	[-0.20, +0.26]
PAE frac < 5 A	15/29	0.618	[0.45, 0.78]	0.1593	0.4414	+0.062	[-0.16, +0.29]
contact prob. sum	15/29	0.649	[0.47, 0.81]	0.1085	0.4414	+0.093	[-0.08, +0.27]
contact prob. density	15/29	0.567	[0.39, 0.74]	0.4741	0.5594	+0.010	[-0.21, +0.23]
contact prob. max	15/29	0.660	[0.49, 0.82]	0.0815	0.4414	+0.103	[-0.10, +0.32]
contacts > 0.5	15/29	0.609	[0.45, 0.77]	0.2016	0.4414	+0.053	[-0.14, +0.26]
contact particip. ratio	15/29	0.620	[0.44, 0.79]	0.2051	0.4414	+0.063	[-0.12, +0.24]
mean pLDDT	15/29	0.552	[0.37, 0.72]	0.5867	0.5867	-0.005	[-0.26, +0.25]

By negative type / vs noncognate (29 positive / 12 negative, length alone 0.520, composition 0.690)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	29/12	0.750	[0.55, 0.92]	0.0132	0.0264	+0.060	[-0.21, +0.35]
chain-pair ipTM	29/12	0.750	[0.55, 0.92]	0.0132	0.0264	+0.060	[-0.21, +0.35]
ranking score	29/12	0.731	[0.56, 0.88]	0.0216	0.0346	+0.042	[-0.23, +0.33]
pTM	29/12	0.606	[0.41, 0.79]	0.2963	0.3386	-0.083	[-0.38, +0.23]
fraction disordered	29/12	0.428	[0.24, 0.62]	0.4777	0.5095	-0.261	[-0.53, +0.03]
chain-pair PAE min	29/12	0.828	[0.68, 0.95]	0.0006	0.0024	+0.138	[-0.09, +0.38]
interchain PAE mean	29/12	0.710	[0.51, 0.89]	0.0365	0.0487	+0.020	[-0.23, +0.28]
interchain PAE p05	29/12	0.741	[0.55, 0.90]	0.0154	0.0274	+0.052	[-0.21, +0.32]
PAE frac < 10 A	29/12	0.713	[0.52, 0.88]	0.0300	0.0436	+0.023	[-0.24, +0.31]
PAE frac < 5 A	29/12	0.764	[0.60, 0.90]	0.0045	0.0144	+0.075	[-0.17, +0.34]
contact prob. sum	29/12	0.874	[0.75, 0.97]	<0.0001	0.0016	+0.184	[+0.00, +0.40]
contact prob. density	29/12	0.763	[0.58, 0.92]	0.0073	0.0195	+0.073	[-0.17, +0.32]
contact prob. max	29/12	0.852	[0.72, 0.96]	0.0002	0.0016	+0.162	[-0.04, +0.39]
contacts > 0.5	29/12	0.848	[0.75, 0.93]	0.0003	0.0016	+0.158	[-0.03, +0.38]
contact particip. ratio	29/12	0.690	[0.51, 0.85]	0.0596	0.0733	+0.000	[-0.19, +0.19]
mean pLDDT	29/12	0.503	[0.32, 0.69]	0.9869	0.9869	-0.187	[-0.47, +0.11]

By negative type / vs cross_kingdom (29 positive / 5 negative, length alone 0.407, composition 0.883)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	29/5	0.910	[0.79, 1.00]	0.0016	0.0048	+0.028	[-0.08, +0.15]
chain-pair ipTM	29/5	0.910	[0.79, 1.00]	0.0016	0.0048	+0.028	[-0.08, +0.15]
ranking score	29/5	0.721	[0.55, 0.88]	0.1220	0.1301	-0.162	[-0.34, +0.02]
pTM	29/5	0.821	[0.64, 0.97]	0.0189	0.0233	-0.062	[-0.23, +0.11]
fraction disordered	29/5	0.810	[0.72, 0.90]	0.0263	0.0301	-0.072	[-0.22, +0.11]
chain-pair PAE min	29/5	0.924	[0.82, 1.00]	0.0018	0.0048	+0.041	[-0.08, +0.18]
interchain PAE mean	29/5	0.876	[0.67, 1.00]	0.0059	0.0105	-0.007	[-0.18, +0.15]
interchain PAE p05	29/5	0.910	[0.78, 1.00]	0.0018	0.0048	+0.028	[-0.10, +0.17]
PAE frac < 10 A	29/5	0.879	[0.79, 0.95]	0.0041	0.0094	-0.003	[-0.13, +0.14]
PAE frac < 5 A	29/5	0.828	[0.74, 0.91]	0.0182	0.0233	-0.055	[-0.19, +0.10]
contact prob. sum	29/5	0.934	[0.82, 1.00]	0.0005	0.0040	+0.052	[-0.03, +0.16]
contact prob. density	29/5	0.869	[0.66, 1.00]	0.0054	0.0105	-0.014	[-0.18, +0.13]
contact prob. max	29/5	0.945	[0.85, 1.00]	0.0005	0.0040	+0.062	[-0.02, +0.18]
contacts > 0.5	29/5	0.862	[0.78, 0.93]	0.0070	0.0112	-0.021	[-0.14, +0.13]
contact particip. ratio	29/5	0.845	[0.58, 1.00]	0.0098	0.0143	-0.038	[-0.23, +0.12]
mean pLDDT	29/5	0.503	[0.31, 0.69]	1.0000	1.0000	-0.379	[-0.59, -0.16]

Appendix B continued

By negative type / vs charge_repulsive (29 positive / 2 negative, length alone 0.759, composition 0.897)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	29/2	0.974	[0.90, 1.00]	n/a	n/a	+0.078	[-0.03, +0.21]
chain-pair ipTM	29/2	0.974	[0.90, 1.00]	n/a	n/a	+0.078	[-0.03, +0.21]
ranking score	29/2	0.966	[0.90, 1.00]	n/a	n/a	+0.069	[-0.07, +0.21]
pTM	29/2	0.931	[0.83, 1.00]	n/a	n/a	+0.034	[-0.10, +0.21]
fraction disordered	29/2	0.284	[0.14, 0.45]	n/a	n/a	-0.612	[-0.83, -0.38]
chain-pair PAE min	29/2	0.931	[0.83, 1.00]	n/a	n/a	+0.034	[-0.12, +0.21]
interchain PAE mean	29/2	0.828	[0.67, 0.97]	n/a	n/a	-0.069	[-0.28, +0.14]
interchain PAE p05	29/2	0.931	[0.83, 1.00]	n/a	n/a	+0.034	[-0.12, +0.21]
PAE frac < 10 Å	29/2	0.879	[0.79, 0.95]	n/a	n/a	-0.017	[-0.14, +0.12]
PAE frac < 5 Å	29/2	0.828	[0.74, 0.91]	n/a	n/a	-0.069	[-0.19, +0.07]
contact prob. sum	29/2	0.983	[0.95, 1.00]	n/a	n/a	+0.086	[-0.02, +0.22]
contact prob. density	29/2	0.966	[0.91, 1.00]	n/a	n/a	+0.069	[-0.03, +0.19]
contact prob. max	29/2	0.983	[0.95, 1.00]	n/a	n/a	+0.086	[-0.02, +0.22]
contacts > 0.5	29/2	0.862	[0.78, 0.93]	n/a	n/a	-0.034	[-0.16, +0.10]
contact particip. ratio	29/2	0.983	[0.95, 1.00]	n/a	n/a	+0.086	[-0.02, +0.22]
mean pLDDT	29/2	0.552	[0.34, 0.76]	n/a	n/a	-0.345	[-0.64, -0.03]

By negative type / vs hard_coiledcoil (29 positive / 6 negative, length alone 0.885, composition 0.655)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	29/6	0.103	[0.01, 0.22]	0.0008	0.0043	-0.552	[-0.79, -0.27]
chain-pair ipTM	29/6	0.103	[0.01, 0.22]	0.0008	0.0043	-0.552	[-0.79, -0.27]
ranking score	29/6	0.095	[0.01, 0.21]	0.0004	0.0043	-0.560	[-0.78, -0.29]
pTM	29/6	0.152	[0.04, 0.29]	0.0048	0.0100	-0.503	[-0.76, -0.21]
fraction disordered	29/6	0.351	[0.17, 0.54]	0.2660	0.2837	-0.305	[-0.61, +0.03]
chain-pair PAE min	29/6	0.201	[0.06, 0.38]	0.0198	0.0264	-0.454	[-0.75, -0.11]
interchain PAE mean	29/6	0.161	[0.04, 0.30]	0.0063	0.0112	-0.494	[-0.75, -0.20]
interchain PAE p05	29/6	0.149	[0.03, 0.30]	0.0039	0.0100	-0.506	[-0.77, -0.18]
PAE frac < 10 Å	29/6	0.155	[0.03, 0.29]	0.0050	0.0100	-0.500	[-0.76, -0.21]
PAE frac < 5 Å	29/6	0.190	[0.07, 0.33]	0.0146	0.0212	-0.466	[-0.74, -0.16]
contact prob. sum	29/6	0.586	[0.41, 0.76]	0.5364	0.5364	-0.069	[-0.26, +0.12]
contact prob. density	29/6	0.213	[0.08, 0.37]	0.0269	0.0331	-0.443	[-0.68, -0.16]
contact prob. max	29/6	0.129	[0.03, 0.25]	0.0023	0.0074	-0.526	[-0.78, -0.24]
contacts > 0.5	29/6	0.184	[0.06, 0.33]	0.0128	0.0205	-0.471	[-0.70, -0.21]
contact particip. ratio	29/6	0.787	[0.63, 0.93]	0.0294	0.0336	+0.132	[-0.04, +0.32]
mean pLDDT	29/6	0.103	[0.01, 0.22]	0.0014	0.0056	-0.552	[-0.79, -0.28]

By negative type / vs homo_negative (29 positive / 14 negative, length alone 0.734, composition 0.704)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	29/14	0.587	[0.40, 0.76]	0.3634	0.5285	-0.117	[-0.34, +0.10]
chain-pair ipTM	29/14	0.587	[0.40, 0.76]	0.3634	0.5285	-0.117	[-0.34, +0.10]
ranking score	29/14	0.603	[0.43, 0.77]	0.2910	0.5173	-0.101	[-0.35, +0.16]
pTM	29/14	0.559	[0.38, 0.73]	0.5387	0.7121	-0.145	[-0.39, +0.10]
fraction disordered	29/14	0.534	[0.37, 0.70]	0.7198	0.7678	-0.170	[-0.39, +0.07]
chain-pair PAE min	29/14	0.754	[0.60, 0.89]	0.0081	0.1192	+0.049	[-0.14, +0.24]
interchain PAE mean	29/14	0.461	[0.28, 0.65]	0.6885	0.7678	-0.244	[-0.49, +0.01]
interchain PAE p05	29/14	0.554	[0.37, 0.72]	0.5785	0.7121	-0.150	[-0.39, +0.09]
PAE frac < 10 Å	29/14	0.674	[0.49, 0.84]	0.0641	0.1709	-0.031	[-0.26, +0.20]
PAE frac < 5 Å	29/14	0.690	[0.53, 0.83]	0.0361	0.1444	-0.015	[-0.24, +0.21]
contact prob. sum	29/14	0.635	[0.47, 0.80]	0.1594	0.3643	-0.069	[-0.20, +0.06]
contact prob. density	29/14	0.379	[0.21, 0.57]	0.2110	0.4220	-0.325	[-0.56, -0.08]
contact prob. max	29/14	0.712	[0.55, 0.86]	0.0270	0.1440	+0.007	[-0.18, +0.20]
contacts > 0.5	29/14	0.727	[0.56, 0.87]	0.0149	0.1192	+0.022	[-0.17, +0.21]
contact particip. ratio	29/14	0.313	[0.16, 0.49]	0.0481	0.1539	-0.392	[-0.57, -0.20]
mean pLDDT	29/14	0.522	[0.34, 0.69]	0.8228	0.8228	-0.182	[-0.46, +0.10]

Appendix B continued

By negative type / vs scramble (29 positive / 9 negative, length alone 0.626)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	29/9	0.667	[0.44, 0.87]	0.1475	0.2622	--	--
chain-pair ipTM	29/9	0.667	[0.44, 0.87]	0.1475	0.2622	--	--
ranking score	29/9	0.745	[0.53, 0.93]	0.0265	0.0728	--	--
pTM	29/9	0.630	[0.41, 0.83]	0.2540	0.3386	--	--
fraction disordered	29/9	0.446	[0.25, 0.65]	0.6386	0.6903	--	--
chain-pair PAE min	29/9	0.816	[0.67, 0.93]	0.0033	0.0264	--	--
interchain PAE mean	29/9	0.586	[0.36, 0.80]	0.4557	0.5608	--	--
interchain PAE p05	29/9	0.648	[0.43, 0.85]	0.1930	0.2807	--	--
PAE frac < 10 A	29/9	0.674	[0.45, 0.87]	0.1184	0.2622	--	--
PAE frac < 5 A	29/9	0.766	[0.61, 0.90]	0.0103	0.0412	--	--
contact prob. sum	29/9	0.745	[0.57, 0.89]	0.0273	0.0728	--	--
contact prob. density	29/9	0.548	[0.30, 0.78]	0.6903	0.6903	--	--
contact prob. max	29/9	0.814	[0.67, 0.93]	0.0029	0.0264	--	--
contacts > 0.5	29/9	0.778	[0.62, 0.90]	0.0095	0.0412	--	--
contact particip. ratio	29/9	0.550	[0.31, 0.79]	0.6738	0.6903	--	--
mean pLDDT	29/9	0.648	[0.40, 0.87]	0.1891	0.2807	--	--

By negative evidence / vs demonstrated (29 positive / 5 negative, length alone 0.855, composition 0.697)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	29/5	0.345	[0.08, 0.63]	0.2816	0.5807	-0.352	[-0.66, -0.01]
chain-pair ipTM	29/5	0.345	[0.08, 0.63]	0.2816	0.5807	-0.352	[-0.66, -0.01]
ranking score	29/5	0.345	[0.08, 0.65]	0.2904	0.5807	-0.352	[-0.69, +0.06]
pTM	29/5	0.321	[0.10, 0.56]	0.2183	0.5807	-0.376	[-0.66, -0.05]
fraction disordered	29/5	0.428	[0.23, 0.63]	0.6182	0.8098	-0.269	[-0.56, +0.05]
chain-pair PAE min	29/5	0.414	[0.13, 0.72]	0.5488	0.8098	-0.283	[-0.62, +0.08]
interchain PAE mean	29/5	0.228	[0.08, 0.39]	0.0569	0.5807	-0.469	[-0.74, -0.14]
interchain PAE p05	29/5	0.286	[0.09, 0.52]	0.1365	0.5807	-0.410	[-0.70, -0.08]
PAE frac < 10 A	29/5	0.441	[0.12, 0.77]	0.7050	0.8098	-0.255	[-0.61, +0.11]
PAE frac < 5 A	29/5	0.448	[0.16, 0.74]	0.7218	0.8098	-0.248	[-0.58, +0.10]
contact prob. sum	29/5	0.559	[0.37, 0.74]	0.7017	0.8098	-0.138	[-0.33, +0.06]
contact prob. density	29/5	0.255	[0.10, 0.43]	0.0854	0.5807	-0.441	[-0.74, -0.09]
contact prob. max	29/5	0.400	[0.09, 0.72]	0.4960	0.8098	-0.297	[-0.63, +0.07]
contacts > 0.5	29/5	0.455	[0.15, 0.77]	0.7592	0.8098	-0.241	[-0.57, +0.12]
contact particip. ratio	29/5	0.531	[0.23, 0.83]	0.8494	0.8494	-0.166	[-0.52, +0.19]
mean pLDDT	29/5	0.303	[0.07, 0.58]	0.1660	0.5807	-0.393	[-0.71, +0.01]

By negative evidence / vs demonstrated+matched_internal (29 positive / 7 negative, length alone 0.734, composition 0.478)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	29/7	0.286	[0.13, 0.46]	0.0851	0.2269	-0.192	[-0.51, +0.14]
chain-pair ipTM	29/7	0.286	[0.13, 0.46]	0.0851	0.2269	-0.192	[-0.51, +0.14]
ranking score	29/7	0.411	[0.20, 0.63]	0.4835	0.6235	-0.067	[-0.42, +0.31]
pTM	29/7	0.214	[0.08, 0.37]	0.0190	0.1716	-0.264	[-0.54, +0.03]
fraction disordered	29/7	0.254	[0.09, 0.45]	0.0429	0.1716	-0.224	[-0.47, +0.04]
chain-pair PAE min	29/7	0.581	[0.40, 0.76]	0.5314	0.6235	+0.103	[-0.18, +0.37]
interchain PAE mean	29/7	0.246	[0.10, 0.40]	0.0395	0.1716	-0.232	[-0.54, +0.08]
interchain PAE p05	29/7	0.308	[0.15, 0.48]	0.1224	0.2545	-0.170	[-0.47, +0.14]
PAE frac < 10 A	29/7	0.325	[0.15, 0.52]	0.1591	0.2545	-0.153	[-0.46, +0.16]
PAE frac < 5 A	29/7	0.424	[0.20, 0.66]	0.5455	0.6235	-0.054	[-0.40, +0.30]
contact prob. sum	29/7	0.685	[0.52, 0.84]	0.1351	0.2545	+0.207	[+0.01, +0.40]
contact prob. density	29/7	0.320	[0.15, 0.52]	0.1481	0.2545	-0.158	[-0.49, +0.17]
contact prob. max	29/7	0.515	[0.33, 0.70]	0.9140	0.9140	+0.037	[-0.25, +0.33]
contacts > 0.5	29/7	0.554	[0.34, 0.77]	0.6583	0.7022	+0.076	[-0.20, +0.35]
contact particip. ratio	29/7	0.640	[0.45, 0.81]	0.2680	0.3898	+0.163	[-0.09, +0.38]
mean pLDDT	29/7	0.236	[0.09, 0.40]	0.0288	0.1716	-0.241	[-0.53, +0.07]

● PASS

Appendix B continued

By negative evidence / vs gold (both demonstrated grades) (29 positive / 12 negative, length alone 0.784, composition 0.569)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	29/12	0.310	[0.15, 0.49]	0.0628	0.1435	-0.259	[-0.53, +0.03]
chain-pair ipTM	29/12	0.310	[0.15, 0.49]	0.0628	0.1435	-0.259	[-0.53, +0.03]
ranking score	29/12	0.384	[0.21, 0.57]	0.2584	0.3758	-0.185	[-0.48, +0.12]
pTM	29/12	0.259	[0.12, 0.42]	0.0141	0.1024	-0.310	[-0.57, -0.04]
fraction disordered	29/12	0.326	[0.17, 0.50]	0.0789	0.1578	-0.243	[-0.48, +0.02]
chain-pair PAE min	29/12	0.511	[0.33, 0.69]	0.9216	0.9216	-0.057	[-0.34, +0.23]
interchain PAE mean	29/12	0.239	[0.10, 0.39]	0.0079	0.1024	-0.330	[-0.60, -0.04]
interchain PAE p05	29/12	0.299	[0.15, 0.47]	0.0459	0.1435	-0.270	[-0.54, +0.01]
PAE frac < 10 A	29/12	0.374	[0.19, 0.57]	0.2062	0.3299	-0.195	[-0.48, +0.09]
PAE frac < 5 A	29/12	0.434	[0.24, 0.63]	0.5138	0.6324	-0.135	[-0.43, +0.16]
contact prob. sum	29/12	0.632	[0.46, 0.80]	0.1982	0.3299	+0.063	[-0.11, +0.25]
contact prob. density	29/12	0.293	[0.15, 0.45]	0.0397	0.1435	-0.276	[-0.55, +0.01]
contact prob. max	29/12	0.467	[0.28, 0.65]	0.7524	0.8599	-0.102	[-0.37, +0.18]
contacts > 0.5	29/12	0.513	[0.32, 0.70]	0.8965	0.9216	-0.056	[-0.31, +0.20]
contact particip. ratio	29/12	0.595	[0.41, 0.77]	0.3621	0.4828	+0.026	[-0.20, +0.25]
mean pLDDT	29/12	0.264	[0.11, 0.43]	0.0192	0.1024	-0.305	[-0.57, -0.01]

By negative evidence / vs inferred (29 positive / 11 negative, length alone 0.567, composition 0.868)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	29/11	0.882	[0.75, 0.99]	0.0002	0.0006	+0.014	[-0.09, +0.12]
chain-pair ipTM	29/11	0.882	[0.75, 0.99]	0.0002	0.0006	+0.014	[-0.09, +0.12]
ranking score	29/11	0.749	[0.59, 0.89]	0.0147	0.0181	-0.119	[-0.29, +0.04]
pTM	29/11	0.840	[0.70, 0.95]	0.0006	0.0011	-0.028	[-0.18, +0.11]
fraction disordered	29/11	0.672	[0.50, 0.82]	0.0829	0.0947	-0.196	[-0.39, +0.00]
chain-pair PAE min	29/11	0.887	[0.77, 0.98]	0.0002	0.0006	+0.019	[-0.10, +0.13]
interchain PAE mean	29/11	0.784	[0.62, 0.92]	0.0048	0.0064	-0.085	[-0.25, +0.06]
interchain PAE p05	29/11	0.843	[0.71, 0.95]	0.0005	0.0010	-0.025	[-0.16, +0.11]
PAE frac < 10 A	29/11	0.840	[0.72, 0.93]	0.0003	0.0008	-0.028	[-0.14, +0.09]
PAE frac < 5 A	29/11	0.803	[0.69, 0.90]	0.0020	0.0032	-0.066	[-0.19, +0.07]
contact prob. sum	29/11	0.839	[0.68, 0.96]	0.0004	0.0009	-0.030	[-0.13, +0.08]
contact prob. density	29/11	0.790	[0.63, 0.92]	0.0035	0.0051	-0.078	[-0.25, +0.07]
contact prob. max	29/11	0.900	[0.79, 0.97]	0.0002	0.0006	+0.031	[-0.07, +0.14]
contacts > 0.5	29/11	0.850	[0.76, 0.93]	0.0002	0.0006	-0.019	[-0.13, +0.11]
contact particip. ratio	29/11	0.638	[0.39, 0.86]	0.1830	0.1952	-0.230	[-0.45, -0.02]
mean pLDDT	29/11	0.567	[0.39, 0.74]	0.5253	0.5253	-0.301	[-0.53, -0.06]

By negative evidence / vs absence_only (29 positive / 12 negative, length alone 0.520, composition 0.690)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	29/12	0.750	[0.55, 0.92]	0.0132	0.0264	+0.060	[-0.21, +0.35]
chain-pair ipTM	29/12	0.750	[0.55, 0.92]	0.0132	0.0264	+0.060	[-0.21, +0.35]
ranking score	29/12	0.731	[0.56, 0.88]	0.0216	0.0346	+0.042	[-0.23, +0.33]
pTM	29/12	0.606	[0.41, 0.79]	0.2963	0.3386	-0.083	[-0.38, +0.23]
fraction disordered	29/12	0.428	[0.24, 0.62]	0.4777	0.5095	-0.261	[-0.53, +0.03]
chain-pair PAE min	29/12	0.828	[0.68, 0.95]	0.0006	0.0024	+0.138	[-0.09, +0.38]
interchain PAE mean	29/12	0.710	[0.51, 0.89]	0.0365	0.0487	+0.020	[-0.23, +0.28]
interchain PAE p05	29/12	0.741	[0.55, 0.90]	0.0154	0.0274	+0.052	[-0.21, +0.32]
PAE frac < 10 A	29/12	0.713	[0.52, 0.88]	0.0300	0.0436	+0.023	[-0.24, +0.31]
PAE frac < 5 A	29/12	0.764	[0.60, 0.90]	0.0045	0.0144	+0.075	[-0.17, +0.34]
contact prob. sum	29/12	0.874	[0.75, 0.97]	<0.0001	0.0016	+0.184	[+0.00, +0.40]
contact prob. density	29/12	0.763	[0.58, 0.92]	0.0073	0.0195	+0.073	[-0.17, +0.32]
contact prob. max	29/12	0.852	[0.72, 0.96]	0.0002	0.0016	+0.162	[-0.04, +0.39]
contacts > 0.5	29/12	0.848	[0.75, 0.93]	0.0003	0.0016	+0.158	[-0.03, +0.38]
contact particip. ratio	29/12	0.690	[0.51, 0.85]	0.0596	0.0733	+0.000	[-0.19, +0.19]
mean pLDDT	29/12	0.503	[0.32, 0.69]	0.9869	0.9869	-0.187	[-0.47, +0.11]

Appendix B continued

By negative evidence / vs uncertain (29 positive / 2 negative, length alone 0.259, composition 0.638)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI	
ipTM	29/2	0.845	[0.67, 0.98]	n/a	n/a	+0.207	[+0.03, +0.39]	● PASS
chain-pair ipTM	29/2	0.845	[0.67, 0.98]	n/a	n/a	+0.207	[+0.03, +0.39]	● PASS
ranking score	29/2	0.802	[0.55, 1.00]	n/a	n/a	+0.164	[-0.07, +0.38]	
pTM	29/2	0.716	[0.41, 0.95]	n/a	n/a	+0.078	[-0.34, +0.47]	
fraction disordered	29/2	0.621	[0.31, 0.86]	n/a	n/a	-0.017	[-0.48, +0.41]	
chain-pair PAE min	29/2	0.914	[0.79, 1.00]	n/a	n/a	+0.276	[+0.07, +0.48]	● PASS
interchain PAE mean	29/2	0.897	[0.76, 1.00]	n/a	n/a	+0.259	[+0.00, +0.52]	
interchain PAE p05	29/2	0.914	[0.79, 1.00]	n/a	n/a	+0.276	[+0.03, +0.52]	● PASS
PAE frac < 10 Å	29/2	0.879	[0.79, 0.95]	n/a	n/a	+0.241	[+0.03, +0.45]	● PASS
PAE frac < 5 Å	29/2	0.828	[0.74, 0.91]	n/a	n/a	+0.190	[-0.02, +0.41]	
contact prob. sum	29/2	0.741	[0.59, 0.90]	n/a	n/a	+0.103	[-0.07, +0.28]	
contact prob. density	29/2	0.759	[0.59, 0.90]	n/a	n/a	+0.121	[-0.14, +0.38]	
contact prob. max	29/2	0.828	[0.69, 0.97]	n/a	n/a	+0.190	[-0.02, +0.41]	
contacts > 0.5	29/2	0.862	[0.78, 0.93]	n/a	n/a	+0.224	[+0.02, +0.45]	● PASS
contact particip. ratio	29/2	0.086	[0.00, 0.21]	n/a	n/a	-0.552	[-0.74, -0.34]	
mean pLDDT	29/2	0.741	[0.45, 0.97]	n/a	n/a	+0.103	[-0.31, +0.48]	

By negative evidence / vs weak (inferred + absence_only) (29 positive / 23 negative, length alone 0.543, composition 0.775)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI	
ipTM	29/23	0.813	[0.68, 0.93]	<0.0001	0.0002	+0.038	[-0.13, +0.21]	
chain-pair ipTM	29/23	0.813	[0.68, 0.93]	<0.0001	0.0002	+0.038	[-0.13, +0.21]	
ranking score	29/23	0.740	[0.60, 0.87]	0.0028	0.0037	-0.035	[-0.22, +0.16]	
pTM	29/23	0.718	[0.57, 0.85]	0.0054	0.0066	-0.057	[-0.26, +0.15]	
fraction disordered	29/23	0.545	[0.39, 0.70]	0.5722	0.6104	-0.230	[-0.43, -0.02]	
chain-pair PAE min	29/23	0.856	[0.74, 0.95]	<0.0001	0.0002	+0.081	[-0.07, +0.24]	
interchain PAE mean	29/23	0.745	[0.60, 0.87]	0.0015	0.0022	-0.030	[-0.20, +0.15]	
interchain PAE p05	29/23	0.790	[0.66, 0.90]	<0.0001	0.0002	+0.015	[-0.16, +0.19]	
PAE frac < 10 Å	29/23	0.774	[0.64, 0.89]	0.0005	0.0008	-0.001	[-0.17, +0.17]	
PAE frac < 5 Å	29/23	0.783	[0.67, 0.89]	<0.0001	0.0002	+0.007	[-0.16, +0.18]	
contact prob. sum	29/23	0.857	[0.75, 0.95]	<0.0001	0.0002	+0.082	[-0.04, +0.22]	
contact prob. density	29/23	0.776	[0.64, 0.89]	0.0002	0.0004	+0.001	[-0.16, +0.17]	
contact prob. max	29/23	0.875	[0.77, 0.96]	<0.0001	0.0002	+0.100	[-0.03, +0.24]	
contacts > 0.5	29/23	0.849	[0.76, 0.93]	<0.0001	0.0002	+0.073	[-0.06, +0.22]	
contact particip. ratio	29/23	0.665	[0.51, 0.81]	0.0417	0.0477	-0.110	[-0.27, +0.04]	
mean pLDDT	29/23	0.534	[0.37, 0.69]	0.6844	0.6844	-0.241	[-0.47, -0.00]	

By positive subgroup / Fuzzy heterotypic positives (15 positive / 43 negative, length alone 0.614, composition 0.541)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI	
ipTM	15/43	0.614	[0.45, 0.77]	0.1870	0.3652	+0.073	[-0.16, +0.30]	
chain-pair ipTM	15/43	0.614	[0.45, 0.77]	0.1870	0.3652	+0.073	[-0.16, +0.30]	
ranking score	15/43	0.560	[0.38, 0.73]	0.4910	0.6210	+0.019	[-0.25, +0.28]	
pTM	15/43	0.650	[0.49, 0.80]	0.0838	0.2936	+0.109	[-0.13, +0.35]	
fraction disordered	15/43	0.638	[0.48, 0.78]	0.1101	0.2936	+0.097	[-0.14, +0.33]	
chain-pair PAE min	15/43	0.690	[0.50, 0.86]	0.0297	0.2936	+0.149	[-0.08, +0.38]	
interchain PAE mean	15/43	0.538	[0.36, 0.71]	0.6772	0.7224	-0.003	[-0.27, +0.27]	
interchain PAE p05	15/43	0.578	[0.39, 0.75]	0.3771	0.5485	+0.036	[-0.23, +0.30]	
PAE frac < 10 Å	15/43	0.602	[0.44, 0.76]	0.2176	0.3652	+0.060	[-0.17, +0.29]	
PAE frac < 5 Å	15/43	0.645	[0.49, 0.80]	0.0596	0.2936	+0.104	[-0.13, +0.34]	
contact prob. sum	15/43	0.557	[0.38, 0.73]	0.5045	0.6210	+0.016	[-0.14, +0.18]	
contact prob. density	15/43	0.478	[0.31, 0.64]	0.8085	0.8085	-0.063	[-0.31, +0.18]	
contact prob. max	15/43	0.657	[0.48, 0.83]	0.0707	0.2936	+0.116	[-0.10, +0.33]	
contacts > 0.5	15/43	0.629	[0.48, 0.78]	0.1012	0.2936	+0.088	[-0.10, +0.28]	
contact particip. ratio	15/43	0.452	[0.30, 0.61]	0.5923	0.6770	-0.089	[-0.25, +0.09]	
mean pLDDT	15/43	0.606	[0.44, 0.76]	0.2283	0.3652	+0.065	[-0.20, +0.32]	

Appendix B continued

By positive subgroup / Fuzzy homotypic positives (14 positive / 43 negative, length alone 0.681, composition 0.854)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI	
ipTM	14/43	0.571	[0.42, 0.71]	0.4321	0.4609	-0.283	[-0.49, -0.07]	
chain-pair ipTM	14/43	0.571	[0.42, 0.71]	0.4321	0.4609	-0.283	[-0.49, -0.07]	
ranking score	14/43	0.605	[0.44, 0.76]	0.2483	0.3310	-0.248	[-0.48, -0.02]	
pTM	14/43	0.393	[0.24, 0.55]	0.2324	0.3310	-0.461	[-0.68, -0.22]	
fraction disordered	14/43	0.276	[0.14, 0.44]	0.0081	0.0259	-0.578	[-0.77, -0.36]	
chain-pair PAE min	14/43	0.724	[0.59, 0.85]	0.0115	0.0263	-0.130	[-0.33, +0.06]	
interchain PAE mean	14/43	0.532	[0.38, 0.68]	0.7309	0.7309	-0.322	[-0.54, -0.09]	
interchain PAE p05	14/43	0.593	[0.45, 0.74]	0.3087	0.3799	-0.261	[-0.48, -0.04]	
PAE frac < 10 A	14/43	0.618	[0.47, 0.76]	0.1759	0.2814	-0.236	[-0.45, -0.02]	
PAE frac < 5 A	14/43	0.626	[0.48, 0.77]	0.1211	0.2422	-0.228	[-0.44, -0.01]	
contact prob. sum	14/43	0.962	[0.89, 1.00]	<0.0001	0.0016	+0.108	[+0.02, +0.22]	● PASS
contact prob. density	14/43	0.635	[0.49, 0.78]	0.1371	0.2437	-0.219	[-0.43, -0.01]	
contact prob. max	14/43	0.723	[0.59, 0.85]	0.0109	0.0263	-0.130	[-0.32, +0.06]	
contacts > 0.5	14/43	0.758	[0.61, 0.89]	0.0017	0.0091	-0.096	[-0.30, +0.10]	
contact particip. ratio	14/43	0.786	[0.64, 0.91]	0.0017	0.0091	-0.068	[-0.19, +0.05]	
mean pLDDT	14/43	0.234	[0.09, 0.40]	0.0024	0.0096	-0.620	[-0.85, -0.35]	

By positive subgroup / Mutual-folding (memorised) positives (4 positive / 43 negative, length alone 0.180, composition 0.488)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI	
ipTM	4/43	0.872	[0.75, 0.97]	0.0092	0.0147	+0.384	[+0.21, +0.55]	● PASS
chain-pair ipTM	4/43	0.872	[0.75, 0.97]	0.0092	0.0147	+0.384	[+0.21, +0.55]	● PASS
ranking score	4/43	0.922	[0.82, 0.99]	0.0018	0.0048	+0.433	[+0.27, +0.59]	● PASS
pTM	4/43	0.811	[0.68, 0.92]	0.0388	0.0498	+0.323	[+0.15, +0.49]	● PASS
fraction disordered	4/43	0.576	[0.27, 0.84]	0.6239	0.6239	+0.087	[-0.25, +0.38]	
chain-pair PAE min	4/43	0.951	[0.88, 1.00]	0.0004	0.0032	+0.462	[+0.30, +0.62]	● PASS
interchain PAE mean	4/43	0.919	[0.82, 0.99]	0.0029	0.0058	+0.430	[+0.26, +0.60]	● PASS
interchain PAE p05	4/43	0.942	[0.86, 0.99]	0.0009	0.0036	+0.453	[+0.28, +0.62]	● PASS
PAE frac < 10 A	4/43	0.924	[0.83, 0.99]	0.0009	0.0036	+0.436	[+0.26, +0.60]	● PASS
PAE frac < 5 A	4/43	0.953	[0.88, 1.00]	0.0003	0.0032	+0.465	[+0.31, +0.62]	● PASS
contact prob. sum	4/43	0.808	[0.64, 0.95]	0.0421	0.0498	+0.320	[+0.12, +0.51]	● PASS
contact prob. density	4/43	0.866	[0.72, 0.98]	0.0114	0.0166	+0.378	[+0.17, +0.57]	● PASS
contact prob. max	4/43	0.933	[0.85, 0.99]	0.0015	0.0048	+0.445	[+0.28, +0.60]	● PASS
contacts > 0.5	4/43	0.901	[0.79, 0.98]	0.0023	0.0053	+0.413	[+0.24, +0.58]	● PASS
contact particip. ratio	4/43	0.407	[0.24, 0.58]	0.5738	0.6121	-0.081	[-0.28, +0.13]	
mean pLDDT	4/43	0.802	[0.67, 0.91]	0.0436	0.0498	+0.314	[+0.13, +0.49]	● PASS

By positive subgroup / All 21 positives incl. memorised (33 positive / 43 negative, length alone 0.590, composition 0.667)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	33/43	0.627	[0.50, 0.75]	0.0607	0.1079	-0.041	[-0.22, +0.14]
chain-pair ipTM	33/43	0.627	[0.50, 0.75]	0.0607	0.1079	-0.041	[-0.22, +0.14]
ranking score	33/43	0.623	[0.49, 0.75]	0.0680	0.1088	-0.044	[-0.23, +0.14]
pTM	33/43	0.561	[0.43, 0.69]	0.3623	0.4140	-0.107	[-0.30, +0.09]
fraction disordered	33/43	0.477	[0.35, 0.61]	0.7291	0.7291	-0.191	[-0.38, +0.00]
chain-pair PAE min	33/43	0.736	[0.62, 0.85]	0.0011	0.0059	+0.069	[-0.10, +0.24]
interchain PAE mean	33/43	0.581	[0.45, 0.71]	0.2262	0.2784	-0.086	[-0.27, +0.11]
interchain PAE p05	33/43	0.628	[0.50, 0.75]	0.0542	0.1079	-0.039	[-0.22, +0.16]
PAE frac < 10 A	33/43	0.648	[0.52, 0.77]	0.0259	0.0691	-0.020	[-0.20, +0.16]
PAE frac < 5 A	33/43	0.674	[0.55, 0.79]	0.0054	0.0173	+0.007	[-0.17, +0.19]
contact prob. sum	33/43	0.759	[0.64, 0.87]	0.0003	0.0048	+0.092	[-0.01, +0.20]
contact prob. density	33/43	0.592	[0.46, 0.72]	0.1739	0.2529	-0.076	[-0.25, +0.11]
contact prob. max	33/43	0.719	[0.60, 0.83]	0.0020	0.0080	+0.051	[-0.11, +0.21]
contacts > 0.5	33/43	0.717	[0.60, 0.83]	0.0010	0.0059	+0.050	[-0.11, +0.21]
contact particip. ratio	33/43	0.588	[0.46, 0.71]	0.1950	0.2600	-0.079	[-0.20, +0.04]
mean pLDDT	33/43	0.472	[0.34, 0.61]	0.6806	0.7260	-0.195	[-0.41, +0.02]

Appendix B continued

By positive subgroup / LLPS-reported positives (15 positive / 43 negative, length alone 0.746, composition 0.803)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	15/43	0.446	[0.31, 0.59]	0.5377	0.7817	-0.357	[-0.51, -0.19]
chain-pair ipTM	15/43	0.446	[0.31, 0.59]	0.5377	0.7817	-0.357	[-0.51, -0.19]
ranking score	15/43	0.460	[0.30, 0.62]	0.6537	0.8046	-0.343	[-0.52, -0.16]
pTM	15/43	0.328	[0.19, 0.48]	0.0474	0.1517	-0.475	[-0.68, -0.25]
fraction disordered	15/43	0.309	[0.16, 0.47]	0.0217	0.0868	-0.495	[-0.72, -0.24]
chain-pair PAE min	15/43	0.549	[0.39, 0.71]	0.5787	0.7817	-0.254	[-0.41, -0.10]
interchain PAE mean	15/43	0.397	[0.26, 0.55]	0.2379	0.5437	-0.406	[-0.58, -0.23]
interchain PAE p05	15/43	0.448	[0.31, 0.60]	0.5562	0.7817	-0.355	[-0.53, -0.18]
PAE frac < 10 A	15/43	0.474	[0.34, 0.62]	0.7606	0.8693	-0.329	[-0.49, -0.18]
PAE frac < 5 A	15/43	0.501	[0.37, 0.63]	1.0000	1.0000	-0.302	[-0.46, -0.14]
contact prob. sum	15/43	0.782	[0.59, 0.95]	0.0009	0.0072	-0.021	[-0.13, +0.08]
contact prob. density	15/43	0.485	[0.33, 0.64]	0.8634	0.9210	-0.318	[-0.47, -0.17]
contact prob. max	15/43	0.547	[0.39, 0.70]	0.5862	0.7817	-0.256	[-0.40, -0.12]
contacts > 0.5	15/43	0.604	[0.44, 0.76]	0.1808	0.4821	-0.199	[-0.35, -0.06]
contact particip. ratio	15/43	0.742	[0.58, 0.88]	0.0044	0.0235	-0.061	[-0.19, +0.07]
mean pLDDT	15/43	0.202	[0.09, 0.34]	0.0005	0.0072	-0.602	[-0.83, -0.35]

By positive subgroup / Positives without LLPS report (14 positive / 43 negative, length alone 0.540, composition 0.573)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI	
ipTM	14/43	0.751	[0.61, 0.87]	0.0046	0.0105	+0.178	[-0.06, +0.41]	
chain-pair ipTM	14/43	0.751	[0.61, 0.87]	0.0046	0.0105	+0.178	[-0.06, +0.41]	
ranking score	14/43	0.713	[0.55, 0.85]	0.0184	0.0268	+0.140	[-0.14, +0.39]	
pTM	14/43	0.738	[0.60, 0.86]	0.0081	0.0162	+0.165	[-0.07, +0.39]	
fraction disordered	14/43	0.629	[0.46, 0.79]	0.1505	0.1685	+0.056	[-0.19, +0.30]	
chain-pair PAE min	14/43	0.875	[0.77, 0.96]	<0.0001	0.0008	+0.302	[+0.10, +0.50]	● PASS
interchain PAE mean	14/43	0.683	[0.51, 0.83]	0.0432	0.0576	+0.110	[-0.17, +0.38]	
interchain PAE p05	14/43	0.732	[0.57, 0.87]	0.0095	0.0169	+0.159	[-0.11, +0.42]	
PAE frac < 10 A	14/43	0.754	[0.61, 0.88]	0.0019	0.0061	+0.181	[-0.06, +0.42]	
PAE frac < 5 A	14/43	0.781	[0.64, 0.90]	0.0005	0.0020	+0.208	[-0.05, +0.45]	
contact prob. sum	14/43	0.721	[0.56, 0.86]	0.0131	0.0210	+0.148	[+0.00, +0.30]	● PASS
contact prob. density	14/43	0.627	[0.47, 0.78]	0.1580	0.1685	+0.054	[-0.22, +0.31]	
contact prob. max	14/43	0.841	[0.71, 0.94]	<0.0001	0.0008	+0.268	[+0.06, +0.47]	● PASS
contacts > 0.5	14/43	0.786	[0.65, 0.90]	0.0005	0.0020	+0.213	[+0.01, +0.41]	● PASS
contact particip. ratio	14/43	0.475	[0.33, 0.63]	0.7884	0.7884	-0.098	[-0.26, +0.07]	
mean pLDDT	14/43	0.668	[0.51, 0.81]	0.0637	0.0784	+0.095	[-0.17, +0.36]	

Length-adjusted / 1. positives vs real negatives (adj.) (29 positive / 43 negative, length alone 0.646, composition 0.692)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	29/43	0.650	[0.52, 0.77]	0.0297	0.0432	-0.043	[-0.21, +0.13]
chain-pair ipTM	29/43	0.650	[0.52, 0.77]	0.0297	0.0432	-0.043	[-0.21, +0.13]
ranking score	29/43	0.666	[0.53, 0.79]	0.0168	0.0336	-0.026	[-0.20, +0.15]
pTM	29/43	0.596	[0.46, 0.72]	0.1653	0.2034	-0.096	[-0.28, +0.09]
fraction disordered	29/43	0.500	[0.36, 0.64]	1.0000	1.0000	-0.192	[-0.39, +0.00]
chain-pair PAE min	29/43	0.768	[0.65, 0.88]	0.0003	0.0024	+0.076	[-0.06, +0.21]
interchain PAE mean	29/43	0.650	[0.52, 0.77]	0.0290	0.0432	-0.043	[-0.21, +0.12]
interchain PAE p05	29/43	0.706	[0.58, 0.82]	0.0038	0.0087	+0.014	[-0.14, +0.16]
PAE frac < 10 A	29/43	0.616	[0.48, 0.74]	0.0974	0.1299	-0.076	[-0.26, +0.11]
PAE frac < 5 A	29/43	0.701	[0.58, 0.81]	0.0029	0.0077	+0.009	[-0.17, +0.19]
contact prob. sum	29/43	0.711	[0.58, 0.83]	0.0018	0.0058	+0.019	[-0.11, +0.15]
contact prob. density	29/43	0.761	[0.63, 0.87]	0.0002	0.0024	+0.069	[-0.07, +0.21]
contact prob. max	29/43	0.730	[0.60, 0.84]	0.0015	0.0058	+0.038	[-0.11, +0.19]
contacts > 0.5	29/43	0.717	[0.59, 0.83]	0.0018	0.0058	+0.024	[-0.13, +0.17]
contact particip. ratio	29/43	0.570	[0.44, 0.70]	0.3228	0.3689	-0.122	[-0.27, +0.03]
mean pLDDT	29/43	0.552	[0.41, 0.69]	0.4594	0.4900	-0.140	[-0.34, +0.06]

Appendix B continued

Length-adjusted / 2b. fuzzy positives vs real negatives (adj.) (20 positive / 37 negative, length alone 0.739, composition 0.793)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	20/37	0.655	[0.51, 0.80]	0.0556	0.0890	-0.138	[-0.28, +0.00]
chain-pair ipTM	20/37	0.655	[0.51, 0.80]	0.0556	0.0890	-0.138	[-0.28, +0.00]
ranking score	20/37	0.691	[0.54, 0.83]	0.0194	0.0443	-0.103	[-0.26, +0.05]
pTM	20/37	0.545	[0.39, 0.69]	0.5945	0.6342	-0.249	[-0.43, -0.06]
fraction disordered	20/37	0.423	[0.27, 0.58]	0.3488	0.3986	-0.370	[-0.55, -0.18]
chain-pair PAE min	20/37	0.761	[0.61, 0.89]	0.0010	0.0080	-0.032	[-0.15, +0.09]
interchain PAE mean	20/37	0.645	[0.49, 0.79]	0.0759	0.1104	-0.149	[-0.28, -0.02]
interchain PAE p05	20/37	0.722	[0.58, 0.85]	0.0069	0.0184	-0.072	[-0.19, +0.04]
PAE frac < 10 Å	20/37	0.605	[0.45, 0.75]	0.1977	0.2433	-0.188	[-0.35, -0.03]
PAE frac < 5 Å	20/37	0.678	[0.53, 0.81]	0.0265	0.0530	-0.115	[-0.28, +0.04]
contact prob. sum	20/37	0.751	[0.59, 0.89]	0.0016	0.0085	-0.042	[-0.17, +0.08]
contact prob. density	20/37	0.746	[0.59, 0.89]	0.0026	0.0104	-0.047	[-0.16, +0.07]
contact prob. max	20/37	0.732	[0.58, 0.87]	0.0041	0.0131	-0.061	[-0.17, +0.05]
contacts > 0.5	20/37	0.772	[0.63, 0.90]	0.0005	0.0080	-0.022	[-0.15, +0.10]
contact particip. ratio	20/37	0.618	[0.47, 0.76]	0.1492	0.1989	-0.176	[-0.33, -0.02]
mean pLDDT	20/37	0.504	[0.34, 0.66]	0.9673	0.9673	-0.289	[-0.49, -0.07]

Length-adjusted / 3. positives vs gold negatives (adj.) (29 positive / 12 negative, length alone 0.784, composition 0.569)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	29/12	0.445	[0.24, 0.66]	0.5997	0.8723	-0.124	[-0.40, +0.15]
chain-pair ipTM	29/12	0.445	[0.24, 0.66]	0.5997	0.8723	-0.124	[-0.40, +0.15]
ranking score	29/12	0.520	[0.32, 0.72]	0.8594	0.9167	-0.049	[-0.33, +0.23]
pTM	29/12	0.392	[0.20, 0.59]	0.2875	0.8723	-0.177	[-0.43, +0.08]
fraction disordered	29/12	0.384	[0.21, 0.58]	0.2480	0.8723	-0.185	[-0.42, +0.06]
chain-pair PAE min	29/12	0.612	[0.44, 0.78]	0.2804	0.8723	+0.043	[-0.14, +0.24]
interchain PAE mean	29/12	0.443	[0.24, 0.66]	0.5794	0.8723	-0.126	[-0.40, +0.15]
interchain PAE p05	29/12	0.466	[0.27, 0.67]	0.7361	0.9060	-0.103	[-0.34, +0.14]
PAE frac < 10 Å	29/12	0.500	[0.28, 0.72]	1.0000	1.0000	-0.069	[-0.37, +0.23]
PAE frac < 5 Å	29/12	0.703	[0.49, 0.89]	0.0427	0.4608	+0.134	[-0.15, +0.40]
contact prob. sum	29/12	0.575	[0.41, 0.74]	0.4670	0.8723	+0.006	[-0.22, +0.23]
contact prob. density	29/12	0.690	[0.52, 0.84]	0.0576	0.4608	+0.121	[-0.08, +0.33]
contact prob. max	29/12	0.537	[0.35, 0.72]	0.7260	0.9060	-0.032	[-0.28, +0.21]
contacts > 0.5	29/12	0.579	[0.37, 0.78]	0.4351	0.8723	+0.010	[-0.25, +0.27]
contact particip. ratio	29/12	0.526	[0.34, 0.72]	0.8159	0.9167	-0.043	[-0.30, +0.22]
mean pLDDT	29/12	0.431	[0.23, 0.65]	0.5104	0.8723	-0.138	[-0.41, +0.15]

Length-adjusted / Homotypic only (adj.) (14 positive / 14 negative, length alone 0.760, composition 0.883)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	14/14	0.750	[0.55, 0.91]	0.0246	0.0394	-0.133	[-0.36, +0.08]
chain-pair ipTM	14/14	0.750	[0.55, 0.91]	0.0246	0.0394	-0.133	[-0.36, +0.08]
ranking score	14/14	0.801	[0.62, 0.95]	0.0068	0.0136	-0.082	[-0.29, +0.11]
pTM	14/14	0.566	[0.33, 0.78]	0.5685	0.6064	-0.316	[-0.57, -0.09]
fraction disordered	14/14	0.393	[0.17, 0.62]	0.3439	0.3966	-0.490	[-0.74, -0.23]
chain-pair PAE min	14/14	0.923	[0.81, 0.99]	<0.0001	0.0016	+0.041	[-0.10, +0.20]
interchain PAE mean	14/14	0.730	[0.53, 0.90]	0.0418	0.0608	-0.153	[-0.39, +0.07]
interchain PAE p05	14/14	0.893	[0.76, 0.98]	0.0004	0.0016	+0.010	[-0.13, +0.17]
PAE frac < 10 Å	14/14	0.714	[0.51, 0.89]	0.0552	0.0736	-0.168	[-0.42, +0.07]
PAE frac < 5 Å	14/14	0.862	[0.70, 0.98]	0.0013	0.0035	-0.020	[-0.21, +0.16]
contact prob. sum	14/14	0.903	[0.77, 0.99]	0.0002	0.0016	+0.020	[-0.13, +0.17]
contact prob. density	14/14	0.842	[0.68, 0.96]	0.0012	0.0035	-0.041	[-0.22, +0.14]
contact prob. max	14/14	0.872	[0.72, 0.98]	0.0003	0.0016	-0.010	[-0.15, +0.13]
contacts > 0.5	14/14	0.852	[0.69, 0.97]	0.0016	0.0037	-0.031	[-0.18, +0.11]
contact particip. ratio	14/14	0.469	[0.26, 0.69]	0.8022	0.8022	-0.413	[-0.67, -0.16]
mean pLDDT	14/14	0.607	[0.38, 0.81]	0.3471	0.3966	-0.276	[-0.55, -0.03]

Appendix B continued

Length-adjusted / Heterotypic only (adj.) (15 positive / 29 negative, length alone 0.568, composition 0.556)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	15/29	0.616	[0.43, 0.79]	0.2075	0.3136	+0.060	[-0.15, +0.29]
chain-pair ipTM	15/29	0.616	[0.43, 0.79]	0.2075	0.3136	+0.060	[-0.15, +0.29]
ranking score	15/29	0.611	[0.43, 0.79]	0.2352	0.3136	+0.055	[-0.16, +0.28]
pTM	15/29	0.654	[0.48, 0.81]	0.0932	0.2982	+0.098	[-0.10, +0.31]
fraction disordered	15/29	0.617	[0.44, 0.78]	0.2132	0.3136	+0.061	[-0.17, +0.29]
chain-pair PAE min	15/29	0.680	[0.49, 0.85]	0.0519	0.2308	+0.124	[-0.08, +0.33]
interchain PAE mean	15/29	0.614	[0.43, 0.79]	0.2235	0.3136	+0.057	[-0.15, +0.28]
interchain PAE p05	15/29	0.646	[0.47, 0.81]	0.1128	0.3008	+0.090	[-0.11, +0.31]
PAE frac < 10 Å	15/29	0.593	[0.40, 0.77]	0.3206	0.3494	+0.037	[-0.18, +0.27]
PAE frac < 5 Å	15/29	0.677	[0.48, 0.85]	0.0507	0.2308	+0.121	[-0.10, +0.35]
contact prob. sum	15/29	0.591	[0.41, 0.77]	0.3276	0.3494	+0.034	[-0.18, +0.25]
contact prob. density	15/29	0.671	[0.48, 0.85]	0.0577	0.2308	+0.115	[-0.10, +0.33]
contact prob. max	15/29	0.683	[0.50, 0.85]	0.0450	0.2308	+0.126	[-0.07, +0.34]
contacts > 0.5	15/29	0.597	[0.42, 0.77]	0.3014	0.3494	+0.040	[-0.14, +0.24]
contact particip. ratio	15/29	0.533	[0.36, 0.71]	0.7323	0.7323	-0.023	[-0.25, +0.20]
mean pLDDT	15/29	0.614	[0.44, 0.78]	0.2234	0.3136	+0.057	[-0.17, +0.29]

Appendix C - local PAE, every group

31 groupings x ipSAE and LIS = 62 exploratory cells, in the layout of appendix B. q is Benjamini-Hochberg over the group's 16 panel metrics plus these two. A PASS badge marks a delta interval clear of zero: 7 cells qualify, 2 of them in the memorised-control block.

group	n	ipSAE	q	delta	delta 95% CI	LIS	q	delta	delta 95% CI	
Pre-registered										
1. positives vs real negatives	29/43	0.682	0.0292	-0.010	[-0.18, +0.16]	0.675	0.0324	-0.017	[-0.19, +0.16]	
2a. mutual folding vs hard coiled-coil	4/6	0.667	0.8491	+0.333	[-0.33, +0.92]	0.875	0.8491	+0.542	[+0.17, +0.88]	● PASS
2b. fuzzy positives vs real negatives	20/37	0.642	0.1393	-0.151	[-0.29, -0.01]	0.645	0.1393	-0.149	[-0.28, -0.01]	
3. positives vs gold negatives	29/12	0.497	0.9887	-0.072	[-0.34, +0.20]	0.477	0.9758	-0.092	[-0.37, +0.18]	
4. positives vs scrambles	29/9	0.778	0.0309	--	--	0.785	0.0309	--	--	
By pair type										
Homotypic only	14/14	0.724	0.1533	-0.158	[-0.41, +0.10]	0.702	0.1543	-0.181	[-0.44, +0.07]	
Heterotypic only	15/29	0.711	0.2538	+0.155	[-0.06, +0.37]	0.672	0.3006	+0.116	[-0.10, +0.34]	
By negative type										
noncognate	29/12	0.776	0.0116	+0.086	[-0.14, +0.34]	0.782	0.0116	+0.092	[-0.16, +0.36]	
cross_kingdom	29/5	0.879	0.0096	-0.003	[-0.13, +0.14]	0.897	0.0082	+0.014	[-0.10, +0.16]	
charge_repulsive	29/2	0.879	n<3	-0.017	[-0.14, +0.12]	0.897	n<3	+0.000	[-0.12, +0.14]	
hard_coiledcoil	29/6	0.322	0.1984	-0.333	[-0.61, -0.03]	0.270	0.0954	-0.385	[-0.65, -0.09]	
homo_negative	29/14	0.728	0.0918	+0.023	[-0.18, +0.23]	0.704	0.1080	+0.000	[-0.21, +0.21]	
scramble	29/9	0.778	0.0309	--	--	0.785	0.0309	--	--	
By negative evidence										
demonstrated	29/5	0.545	0.8317	-0.152	[-0.47, +0.18]	0.462	0.8317	-0.234	[-0.53, +0.10]	
demonstrated+matched_internal	29/7	0.463	0.8893	-0.015	[-0.32, +0.29]	0.488	0.9304	+0.010	[-0.31, +0.32]	
gold (both demonstrated grades)	29/12	0.497	0.9887	-0.072	[-0.34, +0.20]	0.477	0.9758	-0.092	[-0.37, +0.18]	
inferred	29/11	0.853	0.0007	-0.016	[-0.12, +0.10]	0.853	0.0007	-0.016	[-0.12, +0.09]	
absence_only	29/12	0.776	0.0116	+0.086	[-0.14, +0.34]	0.782	0.0116	+0.092	[-0.16, +0.36]	
uncertain	29/2	0.879	n<3	+0.241	[+0.03, +0.45]	● PASS	0.897	n<3	[+0.05, +0.47]	● PASS
weak (inferred + absence_only)	29/23	0.813	0.0002	+0.037	[-0.11, +0.20]	0.816	0.0002	+0.040	[-0.11, +0.20]	
By positive subgroup										
Fuzzy heterotypic positives	15/43	0.718	0.1260	+0.177	[-0.03, +0.39]	0.684	0.1782	+0.143	[-0.07, +0.36]	
Fuzzy homotypic positives	14/43	0.643	0.2010	-0.211	[-0.43, -0.00]	0.666	0.1312	-0.188	[-0.40, +0.02]	
Mutual-folding (memorised) positives	4/43	0.942	0.0018	+0.453	[+0.28, +0.61]	● PASS	0.965	0.0018	[+0.33, +0.63]	● PASS
All 21 positives incl. memorised	33/43	0.713	0.0060	+0.046	[-0.12, +0.22]	0.710	0.0058	+0.043	[-0.13, +0.22]	
LLPS-reported positives	15/43	0.501	1.0000	-0.302	[-0.46, -0.15]	0.509	1.0000	-0.295	[-0.45, -0.15]	
Positives without LLPS report	14/43	0.875	0.0004	+0.302	[+0.10, +0.50]	● PASS	0.854	0.0004	[+0.07, +0.48]	● PASS
Length-adjusted										
1. positives vs real negatives (adj.)	29/43	0.700	0.0084	+0.008	[-0.16, +0.18]	0.751	0.0018	+0.059	[-0.10, +0.22]	
2b. fuzzy positives vs real negatives (..	20/37	0.700	0.0315	-0.093	[-0.23, +0.05]	0.738	0.0123	-0.055	[-0.17, +0.06]	
3. positives vs gold negatives (adj.)	29/12	0.566	0.8304	-0.003	[-0.28, +0.27]	0.618	0.8304	+0.049	[-0.21, +0.30]	
Homotypic only (adj.)	14/14	0.821	0.0060	-0.061	[-0.27, +0.15]	0.878	0.0013	-0.005	[-0.18, +0.17]	
Heterotypic only (adj.)	15/29	0.683	0.1731	+0.126	[-0.10, +0.35]	0.708	0.1731	+0.152	[-0.07, +0.37]	

Appendix C continued - local PAE, every job

ID	chains	label	class	res	ipTM	PAE min	PAE < 10 A	ipSAE	LIS
N01	FUS_LC x PTMA_full	negative	noncognate	274	0.04	31.1	0.000	0.000	0.000
N02	HNRNPA1_LC x H10_full	negative	noncognate	329	0.06	15.3	0.000	0.000	0.000
N03	TDP43_CTD x NPM1_acidic	negative	noncognate	269	0.05	28.0	0.000	0.000	0.000
N04	EWSR1_LC x SNCA_full	negative	noncognate	405	0.05	31.1	0.000	0.000	0.000
N05	TAF15_LC x MAX_bHLHZ	negative	noncognate	289	0.20	9.8	0.000	0.010	0.049
N06	DDX4_NIDR x H10_full	negative	noncognate	430	0.07	24.5	0.000	0.000	0.000
N07	CAPRIN1_CT x GCN4_LZ	negative	cross_kingdom	136	0.09	16.8	0.000	0.000	0.000
N08	G3BP1_IDR1 x NPM1_acidic	negative	charge_repulsive	200	0.04	26.8	0.000	0.000	0.000
N09	PTMA_full x NPM1_acidic	negative	charge_repulsive	232	0.05	23.6	0.000	0.000	0.000
N10	FUS_LC x ERD10_full	negative	cross_kingdom	423	0.06	30.1	0.000	0.000	0.000
N11	TDP43_CTD x ERD10_full	negative	cross_kingdom	408	0.07	23.9	0.000	0.000	0.000
N12	NSP1_FG x H10_full	negative	noncognate	369	0.09	22.3	0.000	0.000	0.000
N13	SRSF1_RS x PTMA_full	negative	noncognate	163	0.06	16.5	0.000	0.000	0.000
N14	U170K_RS x NPM1_acidic	negative	noncognate	328	0.15	7.7	0.012	0.013	0.103
N15	FOS_bZIP x MAX_bHLHZ	negative	hard_coiledcoil	143	0.78	0.9	0.835	0.599	0.585
N16	JUN_bZIP x MAX_bHLHZ	negative	hard_coiledcoil	148	0.70	1.9	0.641	0.264	0.372
N17	GCN4_LZ x JUN_bZIP	negative	hard_coiledcoil	100	0.80	1.0	0.847	0.528	0.648
N18	HNRNPA2_LC x ERD10_full	negative	cross_kingdom	420	0.06	24.7	0.000	0.000	0.000
N19	UBQLN2_CT x SNCA_full	negative	noncognate	315	0.04	28.7	0.000	0.000	0.000
N20	CAPRIN1_CT x ERD10_full	negative	cross_kingdom	363	0.05	28.0	0.000	0.000	0.000
N21	PTMA_full x PTMA_full	negative	homo_negative	222	0.06	24.8	0.000	0.000	0.000
N22	H10_full x H10_full	negative	homo_negative	388	0.07	24.7	0.000	0.000	0.000
N23	NPM1_acidic x NPM1_acidic	negative	homo_negative	242	0.07	18.9	0.000	0.000	0.000
N24	ERD10_full x ERD10_full	negative	homo_negative	520	0.09	19.5	0.000	0.000	0.000
N25	G3BP1_IDR1 x G3BP1_IDR1	negative	homo_negative	158	0.05	19.4	0.000	0.000	0.000
N26	NCOA3_ACTR x NCOA3_ACTR	negative	homo_negative	94	0.52	3.1	0.703	0.141	0.381
N27	CBP_NCBD x CBP_NCBD	negative	homo_negative	118	0.46	3.3	0.543	0.158	0.399
N28	FOS_bZIP x FOS_bZIP	negative	homo_negative	124	0.31	9.0	0.002	0.012	0.079
N29	P53_TAD x P53_TAD	negative	homo_negative	122	0.13	10.5	0.000	0.000	0.048
N30	P21_full x P21_full	negative	homo_negative	328	0.24	4.6	0.018	0.023	0.289
N31	EIF4EBP1 x EIF4EBP1	negative	homo_negative	236	0.14	10.6	0.000	0.000	0.043
N32	MDM2_Ndomain x P53_TAD_L22QW23S	negative	motif_dead_mutant	170	0.17	6.8	0.029	0.015	0.134
N33	ESR1_LBD x NCOA2_NRbox2_LXXAA	negative	motif_dead_mutant	257	0.76	3.4	0.546	0.121	0.364
N34	ABL1_SH3 x SH3BP1_PxxP_scr	scramble	scramble	71	0.28	5.9	0.317	0.020	0.221
N35	YWHAZ_1433 x RAF1_S259A	negative	motif_dead_mutant	260	0.70	1.8	0.527	0.557	0.469
N36	CDC20_WD40 x BUB1B_KENAAA	negative	motif_dead_mutant	452	0.55	2.9	0.199	0.433	0.329
N37	ABL1_SH3 x NCOA2_NRbox2	negative	noncognate	70	0.75	1.4	0.982	0.390	0.700
N38	ESR1_LBD x BUB1B_KENbox	negative	noncognate	271	0.47	5.8	0.186	0.060	0.215
P01	PTMA_full x H10_full	positive	fuzzy	305	0.08	13.1	0.000	0.000	0.000
P02	SRSF1_RS x U170K_RS	positive	fuzzy	259	0.08	17.5	0.000	0.000	0.000
P03	G3BP1_IDR1 x G3BP1_RGG	positive	fuzzy	205	0.06	22.0	0.000	0.000	0.000
P04	FMR1_CT x CAPRIN1_CT	positive	fuzzy	291	0.04	29.5	0.000	0.000	0.000
P05	SNCA_full x TAU_2N4R	positive	fuzzy	581	0.07	31.4	0.000	0.000	0.000
P06	NCOA3_ACTR x CBP_NCBD	positive	mutual_folding	106	0.70	1.4	0.837	0.486	0.609
P07	FOS_bZIP x JUN_bZIP	positive	mutual_folding	129	0.74	0.8	0.649	0.575	0.620
P08	FUS_LC x FUS_LC	positive	fuzzy	326	0.15	2.8	0.023	0.030	0.264
P09	HNRNPA1_LC x HNRNPA1_LC	positive	fuzzy	270	0.21	2.4	0.044	0.044	0.277
P10	HNRNPA2_LC x HNRNPA2_LC	positive	fuzzy	320	0.15	2.1	0.026	0.056	0.272
P11	TDP43_CTD x TDP43_CTD	positive	fuzzy	296	0.12	5.8	0.007	0.016	0.151
P12	DDX4_NIDR x DDX4_NIDR	positive	fuzzy	472	0.08	12.8	0.000	0.000	0.000
P13	EWSR1_LC x EWSR1_LC	positive	fuzzy	530	0.13	4.9	0.004	0.021	0.229
P14	TAF15_LC x TAF15_LC	positive	fuzzy	416	0.14	4.6	0.010	0.023	0.200
P15	UBQLN2_CT x UBQLN2_CT	positive	fuzzy	350	0.17	4.7	0.008	0.022	0.257
P16	NSP1_FG x NSP1_FG	positive	fuzzy	350	0.20	10.8	0.000	0.000	0.046
P17	TAU_2N4R x TAU_2N4R	positive	fuzzy	882	0.23	3.0	0.013	0.115	0.274
P18	SNCA_full x SNCA_full	positive	fuzzy	280	0.23	1.8	0.045	0.069	0.345
P19	CAPRIN1_CT x CAPRIN1_CT	positive	fuzzy	206	0.09	8.6	0.002	0.013	0.108
P20	MAX_bHLHZ x MAX_bHLHZ	positive	mutual_folding	162	0.80	1.0	0.822	0.646	0.622
P21	GCN4_LZ x GCN4_LZ	positive	mutual_folding	66	0.73	0.9	0.901	0.438	0.764
P22	TP53_full x MDM2_full	positive	fuzzy	905	0.26	1.1	0.007	0.659	0.560
P23	TP53_full x MDM2_full	positive	fuzzy	884	0.31	1.0	0.008	0.737	0.618
P24	P53_TAD x MDM2_Ndomain	positive	coupled_folding	170	0.20	5.4	0.045	0.019	0.188
P25	ESR1_LBD x NCOA2_NRbox2	positive	coupled_folding	257	0.95	0.9	0.894	0.850	0.725
P26	ABL1_SH3 x SH3BP1_PxxP	positive	coupled_folding	71	0.87	0.8	0.980	0.715	0.826
P27	YWHAZ_1433 x RAF1_pS259	positive	coupled_folding	260	0.73	1.4	0.558	0.639	0.519
P28	CDC20_WD40 x BUB1B_KENbox	positive	coupled_folding	452	0.68	1.1	0.303	0.812	0.508
S01	PTMA_full x H10_full_scr	scramble	scramble	305	0.06	19.8	0.000	0.000	0.000
S03	G3BP1_IDR1 x G3BP1_RGG_scr	scramble	scramble	205	0.09	13.8	0.000	0.000	0.000

Appendix C continued - local PAE, every job

ID	chains	label	class	res	ipTM	PAE min	PAE < 10 A	ipSAE	LIS
S04	FMR1_CT x CAPRIN1_CT_scr	scramble	scramble	291	0.12	12.2	0.000	0.000	0.000
S06	NCOA3_ACTR x CBP_NCBD_scr	scramble	scramble	106	0.46	3.1	0.249	0.119	0.392
S07	FOS_bZIP x JUN_bZIP_scr	scramble	scramble	129	0.40	6.0	0.078	0.017	0.176
S08	FUS_LC x FUS_LC_scr	scramble	scramble	326	0.05	26.8	0.000	0.000	0.000
S11	TDP43_CTD x TDP43_CTD_scr	scramble	scramble	296	0.10	18.6	0.000	0.000	0.000
S12	DDX4_NIDR x DDX4_NIDR_scr	scramble	scramble	472	0.06	30.9	0.000	0.000	0.000
T01	NCOA3_ACTR x CBP_NCBD	positive	mutual_folding	106	0.73	1.3	0.865	0.510	0.624
T02	FOS_bZIP x JUN_bZIP	positive	mutual_folding	129	0.72	0.9	0.634	0.531	0.593
T03	MAX_bHLHZ x MAX_bHLHZ	positive	mutual_folding	162	0.77	1.1	0.793	0.612	0.596
T04	GCN4_LZ x GCN4_LZ	positive	mutual_folding	66	0.68	1.2	0.891	0.333	0.697
T05	FOS_bZIP x MAX_bHLHZ	negative	hard_coiledcoil	143	0.77	0.9	0.788	0.558	0.552
T06	JUN_bZIP x MAX_bHLHZ	negative	hard_coiledcoil	148	0.74	1.7	0.781	0.365	0.466
T07	GCN4_LZ x JUN_bZIP	negative	hard_coiledcoil	100	0.77	1.1	0.820	0.479	0.607
T08	NCOA3_ACTR x NCOA3_ACTR	negative	homo_negative	94	0.48	3.4	0.564	0.111	0.419
T09	CBP_NCBD x CBP_NCBD	negative	homo_negative	118	0.18	9.5	0.002	0.011	0.074
T10	TP53_full x MDM2_full	positive	fuzzy	905	0.26	1.1	0.006	0.640	0.595
T11	FOS_bZIP x FOS_bZIP	negative	homo_negative	124	0.32	9.1	0.001	0.012	0.073

PAE min = chain-pair minimum interchain PAE (A). PAE < 10 A = fraction of interchain pairs under 10 A. ipSAE and LIS as defined on the methods page; 0 means no interchain pair cleared the cutoff.